

Bucket Elevator

Installation and Operation Manual

PNEG-2116

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All information, illustrations, photos, and specifications in this manual are based on the latest information available at the time of publication. The right is reserved to make changes at any time without notice.

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General Information

GSI reserves the right to improve its product whenever possible and practical to do so. We reserve the right to change, improve and modify products at any time without obligation to make changes, improvements and modifications on equipment sold previously.

This manual covers the installation and operation for the Bucket Elevator. This manual provides guidelines for installing the product. You must retain a qualified contractor to provide on-site expertise. GSI IS NOT RESPONSIBLE FOR THE INSTALLATION OF THIS PRODUCT.

DO NOT exceed the elevator's rating. A certified drawing or drawings furnished with the elevator gives its capacity in BPH (Bushels Per Hour) plus the weight per cubic foot and the weight per bushel of the commodity or material the elevator is designed to elevate. The drawing also specifies the operating speed of the elevator belt and other pertinent data. Consult GSI. before making any changes to the elevator or its operating environment; in particular, any change in the speed or power of the elevator drive. Careless changes could result in death or serious injury to people, and/or reduce the performance and service life of the equipment.

Never perform any service on this elevator or any other powered equipment until all power has been shut off and locked out so that it cannot be restored without the consent and knowledge of the person who interrupted power. Power includes electrical, fluid, mechanical (cable, belt, chain, shaft, etc.) or gravity where the load or part of the equipment is suspended. Never add restraint to an explosion vent. If the elevator structure is damaged to the extent the relief panels will not stay in position, make proper repairs. Remember the vents are designed to blow out readily, limiting damage and injury, in case of explosion. Failure to observe all safety precautions, including those dictated by ordinary common sense, can result in death or serious injury to personnel, loss of product (elevated material), and damage or destruction of the equipment.

General Safety Statements

1. The Bucket Elevator is designed and manufactured with operator safety in mind. However, residual hazards remain due to the nature of material handling, and specific material hazards. Use extreme caution at all times.
2. Modifications to equipment may cause extremely dangerous situations that could result in damage to the equipment as well as serious injury or death. Never modify the equipment.
3. GSI recommends that you contact the local power company to have a representative survey the installation to ensure wiring is compatible with their system and adequate power is supplied to the unit.

Replacement Parts

The certified drawings furnished with the elevator list the components, which are likely to require replacement. Replacements for any other components, including structural members can be supplied upon request.

For direct parts orders or requests for technical assistance should be directed to:

GSI
1004 E. Illinois St.
Assumption, IL. 62510
Phone: 1-217-226-4421

Please have available the Model Number and Serial Number of the equipment in question, as well as the location where the equipment is INSTALLED.

Safety Guidelines

Safety guidelines are general-to-specific safety rules that must be followed at all times. This manual is written to help you understand safe operating procedures and problems that can be encountered by the operator and other personnel when using this equipment. Read and save these instructions.

As owner or operator, you are responsible for understanding the requirements, hazards, and precautions that exist and to inform others as required. Unqualified persons must stay out of the work area at all times.

Alterations must not be made to the equipment. Alterations can produce dangerous situations resulting in SERIOUS INJURY or DEATH.

This equipment must be installed in accordance with the current installation codes and applicable regulations, which must be carefully followed in all cases. Authorities having jurisdiction must be consulted before installations are made.

When necessary, you must consider the installation location relative to electrical, fuel and water utilities.

Personnel operating or working around equipment must read this manual. This manual must be delivered with equipment to its owner. Failure to read this manual and its safety instructions is a misuse of the equipment.

ST-0001-4

Cautionary Symbols Definitions

Cautionary symbols appear in this manual and on product decals. The symbols alert the user of potential safety hazards, prohibited activities and mandatory actions. To help you recognize this information, we use the symbols that are defined below.

DANGER — This symbol indicates an imminently hazardous situation which, if not avoided, **will result in serious injury or death.**

WARNING — This symbol indicates a potentially hazardous situation which, if not avoided, **can result in serious injury or death.**

CAUTION — This symbol indicates a potentially hazardous situation which, if not avoided, **can result in minor or moderate injury.**

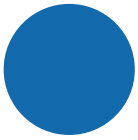
NOTICE — This symbol is used to address practices not related to personal injury.



This symbol indicates a general hazard.



This symbol indicates a prohibited activity.



This symbol indicates a mandatory action.

ST-0005-2

Safety Cautions

Use Personal Protective Equipment

- Use appropriate personal protective equipment:

Eye Protection



Respiratory Protection



Foot Protection



Hearing Protection



Head Protection



Fall Protection



Hand Protection



- Wear clothing appropriate to the job.
- Remove all jewelry.
- Tie long hair up and back.

ST-0004-1

Follow Safety Instructions

- Carefully read all safety messages in this manual and safety signs on your machine. Keep signs in good condition. Replace missing or damaged safety signs. Be sure new equipment components and repair parts include the current safety signs. Replacement safety signs are available from the manufacturer.
- Learn how to operate the machine and how to use controls properly. Do not let anyone operate without instruction.
- If you do not understand any part of this manual or need assistance, contact your dealer.



ST-0002-1

Maintain Equipment and Work Area

- Understand service procedures before doing work. Keep area clean and dry.
- Never service equipment while it is operating. Keep hands, feet, and clothing away from moving parts.
- Keep your equipment in proper working condition. Replace worn or broken parts immediately.



ST-0003-1

Stay Clear of Moving Parts

- Entanglement in rotating sprocket or moving chain will cause serious injury or death.
- Keep all guards and covers in place at all times.
- Lock-out power source before making adjustments, cleaning, or maintaining equipment.



ST-0076-1

Install, Operate, and Maintain Electrical Equipment Properly

- Electrical controls and electrical equipment must be installed by a qualified electrician and must meet the standards set by applicable local codes (National Electrical Code for the US, Canadian Electric Code, or EN60204 along with applicable European Directives for Europe).
- Make sure all equipment is properly grounded.
- Confirm that power supplies are provided at the correct voltage and are of sufficient capacity according to the data provided in this manual and/or on the rating plate.
- Note that supply conductors, connected to the input of the disconnect switch, remain live when the panel is locked out. Lock out and tag out supply conductors at supply position (remote from the equipment) before making adjustments, cleaning, or maintaining equipment.
- Check supply conductors are disconnected by testing with a voltmeter before working on or around them.



ST-0027-5

Sharp Edge Hazard

- This product has sharp edges, which can cause serious injury.
- To avoid injury, handle sharp edges with caution and always use proper protective clothing and equipment.



ST-0036-2

Fall Hazard

- Keep access door closed while on a platform to avoid falls.
- Always use proper personal protective equipment and proper clothing when using equipment. Failure to follow safety precautions can result in severe injury or death.



ST-0042-2

Toxic Fume and Dust Hazard

- Do all work outside or in a well-ventilated area. Dispose of paint and solvent properly.
- Remove paint before welding or heating:
  - — Avoid potentially toxic fumes and dust. Hazardous fumes can be generated when paint is heated by welding, soldering, or using a torch.
  - — If you sand or grind paint, avoid breathing the dust. Wear an approved respirator.
  - — If you use solvent or paint-stripper, remove stripper with soap and water before welding.
  - — Remove solvent or stripper containers and other flammable material from area. Allow fumes to disperse at least 15 minutes before welding or heating.





ST-0043-2

Platform Load Limit

- The platform load limit is 500 LBS (2.25 kN). Do not exceed this weight.
- Excessive load will damage the platform and cause platform to fall. Severe injury or death will result.





ST-0044-2

Stay Clear of Hoisted Equipment

- Always use proper lifting or hoisting equipment when assembling or disassembling equipment.
- Do not walk or stand under hoisted equipment.
- Always use sturdy and stable supports when needed for installation. Not following these safety precautions creates the risk of falling equipment, which can crush personnel and cause serious injury or death.



ST-0047-1

Fall Hazard

- Ladders, stairways and platforms are for use by competent and trained personnel only. Do not allow children or other unauthorized persons to have access to the equipment.
- Access to the equipment must be restricted by the use of security fencing and lockable gates.
- Lower sections of ladders must be fitted with a lockable safety gate to prevent unauthorized access.
- Make sure that hot surfaces have had adequate time to cool before working on or in the equipment.
- Lock out and tag out power supplies and fuel supplies to all equipment.
- Do not attach lifting equipment to ladders or platforms.
- Do not go outside of the safety rails provided on elevated platforms.
- Do not work at heights during high winds, rain, snow, or ice storms.





ST-0056-1

Flying Material Hazard

- Flying material can cause severe eye injury or blindness.
- Wear safety glasses around operating equipment.





ST-0074-1

Stay Clear of Rotating Parts

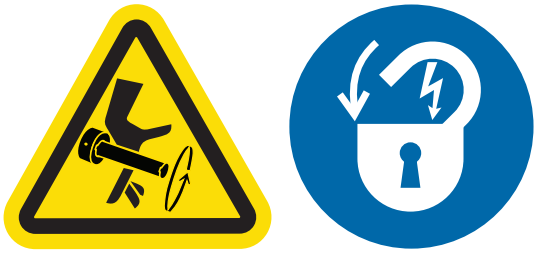
- Do not service equipment while it is in operation.
- Entanglement in rotating parts or exposed belts will cause serious injury or death.
- Keep all shields and covers in place at all times.
- Lock-out power source before making adjustments, cleaning, or maintaining equipment.



ST-0077-1

Stay Clear of Rotating Parts

- Do not service equipment while the equipment is in operation.
- Entanglement in rotating shafts will cause serious injury or death.
- Keep all guards and covers in place at all times.
- Lock-out power source before making adjustments, cleaning, or maintaining equipment.



ST-0096-1

Installing Equipment at Height Hazard

- Risk of physical injury or death.
- Carry out a full risk assessment and develop safe work plans before starting installation.
- Check for and avoid overhead power lines.
- Avoid use of ladders.
- Use safe access equipment such as cherry picker or lift platform.
- Use safety harness and fall restraint.
- Check weight of equipment to be lifted and select appropriate lifting equipment and accessories.
- Restrain and secure all loose parts before lifting.
- Avoid working in high winds.
- Be aware of potential falling objects.
- Wear suitable protective equipment.



ST-0110-1

Safety Sign-Off Sheet

Below is a sign-off sheet that can be used to verify that all personnel have read and understood the safety instructions. This sign-off sheet is provided for your convenience and personal record keeping.

Date	*Employee Name*	*Supervisor Name*

The safety decals on your equipment are safety indicators which must be carefully read and understood by all personnel involved in the installation, operation, service and maintenance of the equipment.
To replace a damaged or missing decal, contact us to receive a free replacement.

GSI Decals
1004 E. Illinois St.
Assumption, IL. 62510
Phone: 1-217-226-4421

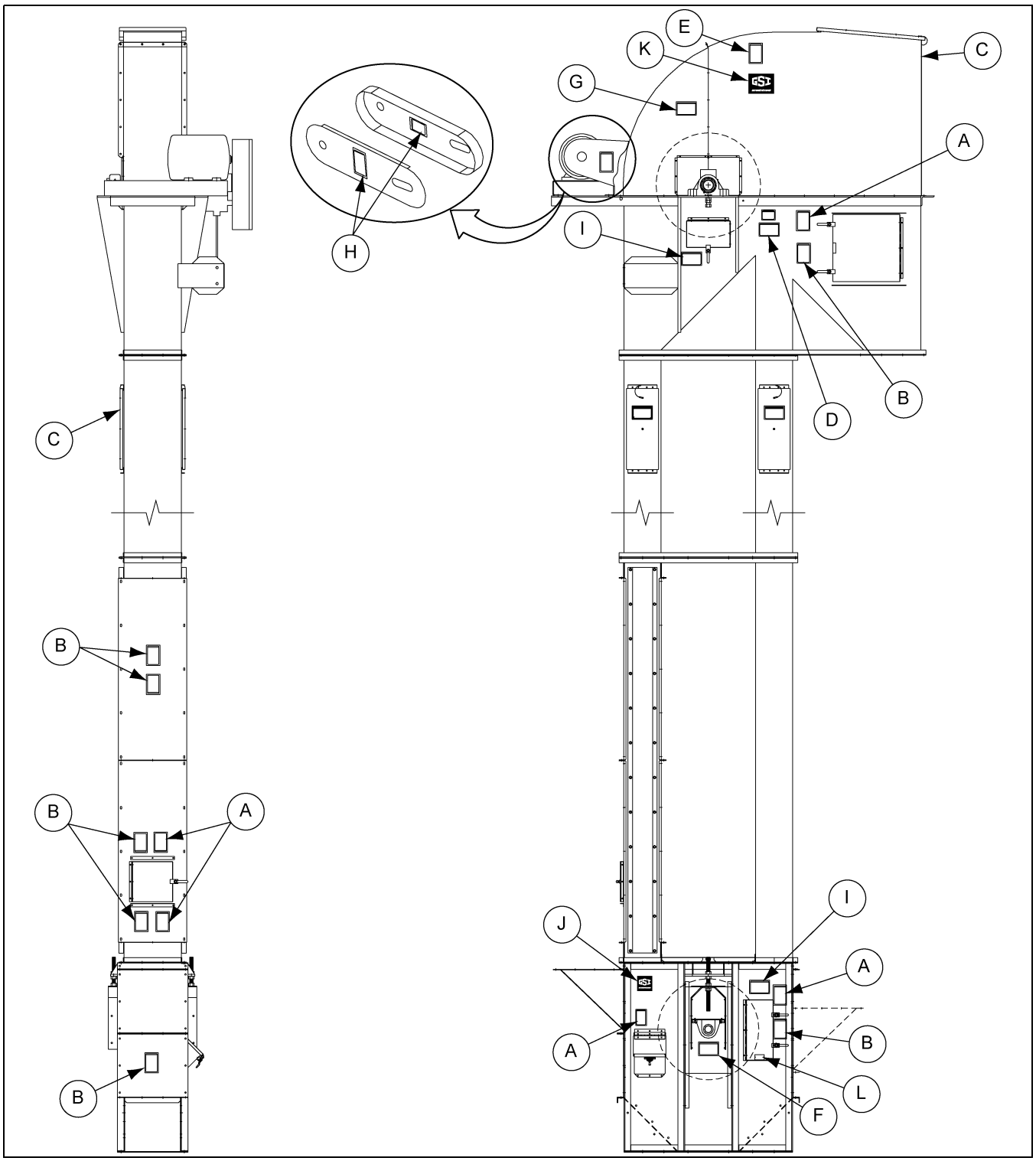
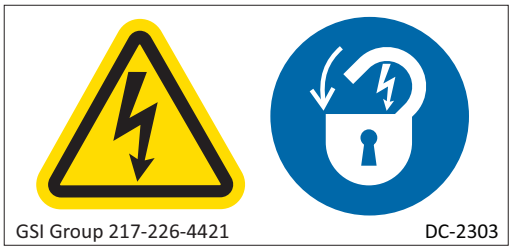
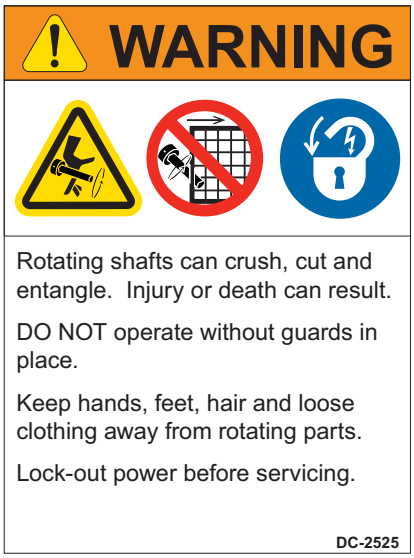


Figure 3A *Bucket Elevator Safety Decals Locations*

NOTE: *These labels must never be removed, tampered with, painted over, or obscured in any way. If labels are damaged or become unreadable, replacement labels are available from GSI.*

Ref #	Decal #	Decals	Description
A	DC-1198		Decal, Warning Eye Protection
B	DC-1199		Decal, Warning Bucket Elevator
C	DC-1377		Decal, Danger Explosion Release
D	DC-1378		Decal, Warning Platform Load Limit
E	DC-1601		Decal, Bucket Elevator 'No Lift'

Ref #	Decal #	Decals	Description
F	DC-1749		Decal, Warning Rotating Shaft
G	DC-2303		Decal, ISO, Electrical Lock Out
H	DC-2468		Decal, Warning Rotating Parts

Ref #	Decal #	Decals	Description
I	DC-2525		Decal, Warning Rotating Shafts
J	DC-2553		Decal, Intersystems Logo
K	DC-2569		Decal, GSI Intersystems Logo 14" x 11"
L	DC-2581		Serial Tag Plate AL 2-3/4" x 1-1/2"

IMPORTANT: *Thoroughly read safety information from Pages 5-16 before beginning installation and start-up.*

Receiving Inspection

1. Carefully inspect the shipment for damage as soon as it is received. Verify that the quantity of parts or packages actually received corresponds to the quantity shown on the packing slip. One or more cartons containing the fasteners required for assembly are included with the shipment. Report any damage or shortage to the delivering carrier as soon as possible.
2. GSI responsibility for damage to the equipment ended with acceptance by the delivering carrier. Refer to the bill of lading for more detailed information.
3. Save all paperwork and documentation furnished with any of the elevator components.

Pre-Installation Preparation

1. The MOST IMPORTANT preparations are retaining a licensed engineer to plan the installation and a qualified millwright or contractor to erect the elevator and the accompanying equipment and structures. Before starting the bucket elevator installation, review this manual, the certified drawing(s) furnished with the equipment, and other applicable documents, including but not limited to, O.S.H.A. Regulations and the National Electrical Code. You should also obtain a copy of “Guidelines for Supporting Outside Elevator Legs”, Bulletin No. F-646-75, available from Mill Mutual Fire Prevention Bureau, North Riverside Plaza, Chicago, IL 60606, as it is referenced throughout the manual.
2. GSI is the vendor of the bucket elevator and certain of its optional accessories only, and does not assume responsibility for the installation.
3. Bucket Elevators are designed to be vertically self-supporting when erected. The elevator does require guying and/or bracing from another structure for horizontal support. The elevator has not been designed to support other equipment such as cleaners, distributors, spouting, etc. Separate structures must be provided for any accessory equipment.

Elevator Foundation

Refer to Mill Mutual Publication No F-646-75. Additionally, in the process of leveling the boot, GSI recommends that the installer plan on shimming the boot around its perimeter and grouting the base plate after the hold down fasteners are in place and tightened.

Elevator Guying or Bracing

Refer to Mill Mutual Publication No F-646-75.

DANGER

Incorrect attachment of guying cables to elevator or lateral bracing from an adjoining structure will result in support failure, collapse of the elevator and death or severe injury to anyone in the area.

Attach guying cables to the elevator only by means of the 4 wire bolt-on brackets as shown in Figure 4A. If bracing from an adjoining structure is used, attach only at flange joints between legging sections.

NEVER attach any bracing directly to the sheet metal trunk. Sheet metal lacks the necessary stiffness and thickness for sound structural joints.

The engineer or contractor supervising elevator installation is ultimately responsible for determining suitable guying and/or bracing methods and materials. Recommended maximum distance between lateral bracing is 20' increments vertically at trunk flange seams to adjoining structure. The following considerations are presented in addition to those set forth in **Mill Mutual Publication No F-646-75**:

1. Consider placing curbs or guards around anchors to prevent impact and damage from area traffic.
2. Guy cables near ground level must be guarded and made visible to prevent injuries.
3. *Figure 4B* shows the proper method for unspooling cable from a reel.

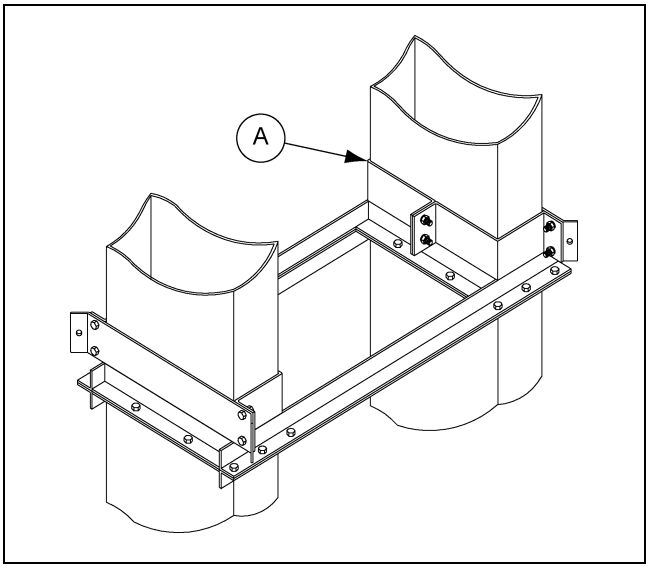


Figure 4A *Guy Bracket Detail*

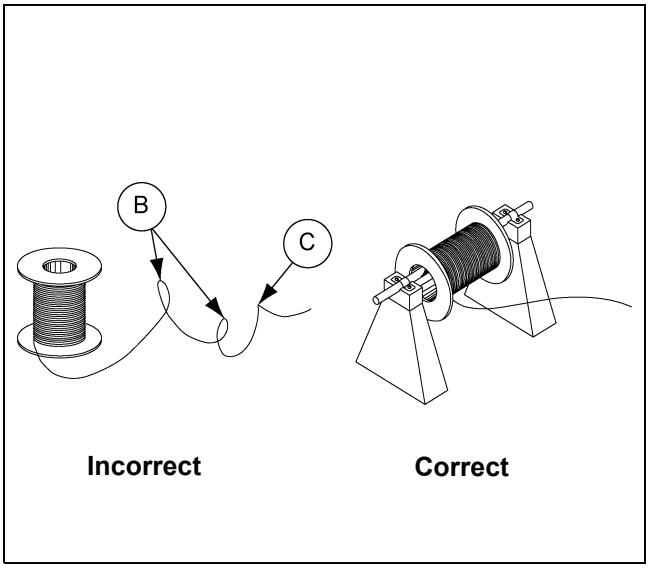


Figure 4B *Proper Cable Unspooling*

Ref #	Description
A	Mount guy brackets atop legging frame to prevent brackets from working down-legging.

Ref #	Description
B	Kinks
C	Dog Leg

DANGER

Excessive guy cable tension will twist, bend and/or collapse the elevator structure, causing severe injury or death. Maintain proper guy cable tension at all times.

Elevator Intake

Several factors should be taken into account when planning the intake to feed material into the elevator:

1. The method and location of intake openings in the boot usually will have been determined when the elevator was ordered. However, if the intake arrangements have or are being changed, several factors must be considered. *(See Figure 4C.)*

   a. **Inlet or Hopper Size:** The size of the inlet effects the capacity and the condition of the elevated material. Too small an inlet means the material must be fed into the boot at excessive speed. The excess velocity of incoming material cause damage, the buckets do not fill properly, and material backs up in the boot, clogging the inlet.

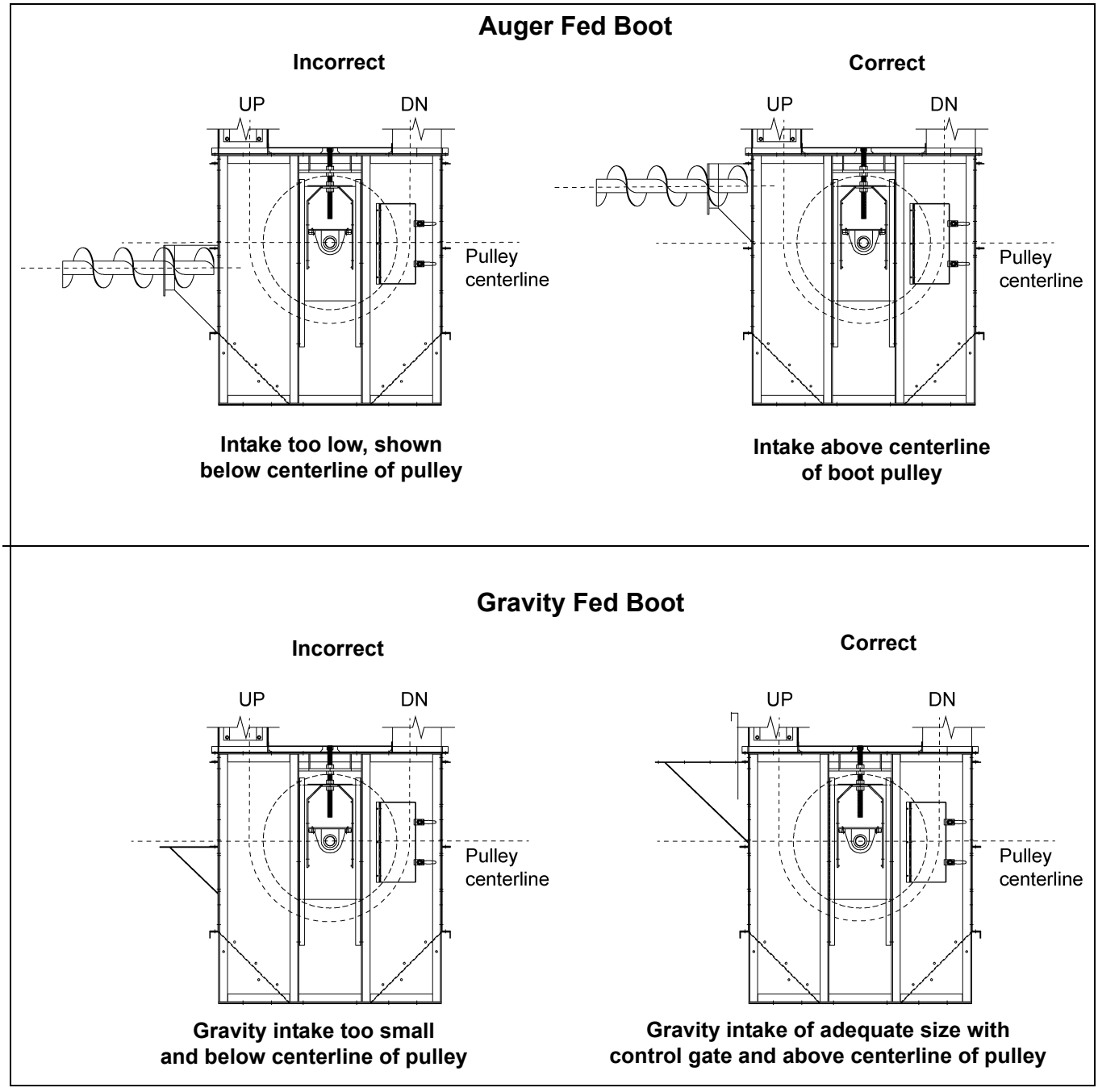


Figure 4C *Boot Intake Conditions*

b. **UP Side Infeed Devices:** The device which feeds the elevator, (auger, belt conveyor, spout, hopper, etc.), must feed the material into the boot ABOVE the centerline of the pulley. The buckets must complete the turn around the pulley and be moving vertically upward, not at a tangent, before encountering material to be elevated. If the buckets ARE NOT moving vertically upward when they encounter the incoming material, material will be forced back into the infeed device or chute. Grinding and churning in the boot area will damage the product and excessive power will be required to operate the system.

c. **DOWN Side Infeed Devices:** Several factors, most importantly the nature of the material being elevated, are important in determining the location of down side or down-leg infeed devices. If a change in infeed arrangement from that shown on the certified drawings is contemplated, consult GSI before proceeding with modifications.

d. **FLOW of material to the infeed Devices:** Every effort should be made when designing the feeding equipment or spouting such that the direction of the flowing material will be perpendicular to the buckets. Reduce or eliminate any side loading of the buckets. Achieving the bucket elevators rated capacity is determined by even and consistant bucket fill.

Elevator Discharge

1. Spouting to carry off the material must be sized so that its capacity equals or exceeds the maximum capacity of the elevator to prevent material plugging in the head or down-legging.
2. The elevator IS NOT designed to support the weight of any accessory equipment. Spouting, cleaners, distributors, etc. must have their own supporting structures.

Positioning Trunk Sections for Ladder

Study the drawings to determine where each trunk section fits into the overall layout of the elevator leg. This is important so that platforms and ladders can be attached to the elevator without having to re-assemble ladder brackets. The final ladder at the head section (at the top of the elevator) must be near the trunk closest to the discharge (the down-leg trunk). *Figure 4I on Page 25* for ladder and cage assembly details.

Elevator Belt Selection

If GSI has supplied the belt, no further action is required at this time. However, if the belt is being obtained independent of the elevator, those ordering the belt must be aware of a number of requirements.

1. Belt length must take into consideration the method of belt splicing to be employed in this installation. Refer to belt splicing *on Page 33* and *Figure 4Q on Page 34* for an overview of the three (3) splicing methods commonly employed in elevator installations. A mechanical splice requires the least extra belt while a butt splice requires the greatest length of extra belt.
2. The pattern, spacing, and size of prepunched holes must be correct for the number, type, and size of the buckets to be attached to the belt. If the belt is purchased with the wrong hole pattern, it will be extremely difficult and expensive to correct the problem in the field. If the hole sizes are too small, they may have to be re-drilled in the field. If the holes are too large, the buckets may have to be re-drilled to accept larger bolts or be replaced. Elevator bolts must fit snugly in the belt.
3. The installer should fabricate spreader bars in advance to avoid delay once belt installation begins. See *View D* of *Figure 4Q on Page 34*.
4. A belt manufacturer or a local belt supplier sales representative can assist in the correct selection and specification of belting and advise in proper installation procedures.

Elevator Erection (Assembly)

Erection of the elevator must not be started until the concrete foundation and the guying anchors (if any are needed) have been completed and developed maximum strength.

IMPORTANT: *No matter who does the installation or what method is employed, the elevator must be plumb. If it is not, it almost certainly cannot be made to operate satisfactorily.*

There are two (2) basic methods of elevator erection. In this manual only the vertical assembly or section-by-section method is considered. GSI does not recommend horizontal assembly wherein entire elevators or multiple sections thereof are assembled horizontally at ground level and then raised to the vertical and hoisted into position. The potential for damage to the elevator structure is too great.

Vertical Assembly

This method could also be termed section-by-section assembly. *(See Figure 4E on Page 22.)* Because of the weight of the elevator's head section, GSI recommends that it always be hoisted in a separate lift and then fastened to the rest of the elevator.

1. Lift the boot onto the foundation. (See Elevator Foundation *on Page 17*.) BE SURE THE BOOT IS POSITIONED CORRECTLY ON THE FOUNDATION AS SHOWN ON THE DRAWINGS.
2. While the hoisting gear is still attached to the boot to prevent its tipping, accurately plumb and level the boot base. Use appropriate fasteners to connect the base securely in position and grout the boot base. Allow the grout to cure before further assembly is attempted.
3. GSI recommends that all joints between elevator sections be weather-proofed by applying mastic compound, especially if the elevator is to remain an outdoor structure. Mastic or caulking compound is to be provided by the user or installer. Do not use weather stripping. Use grade 5 or higher bolts to connect sections.

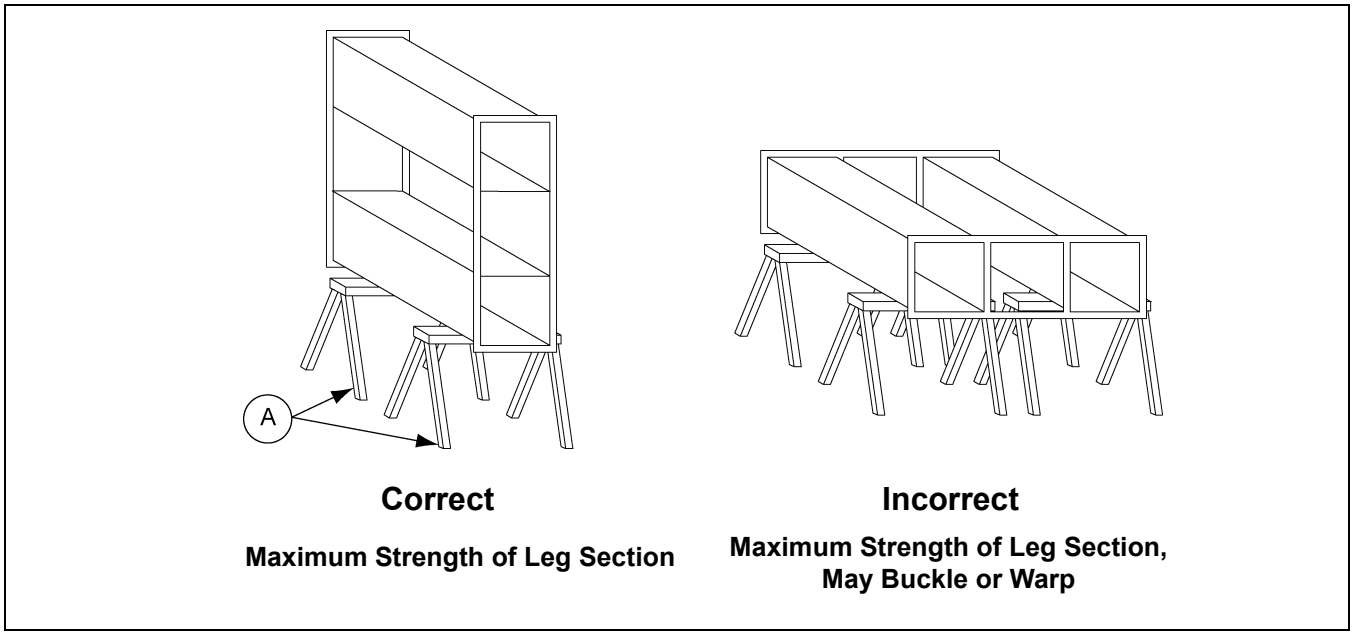


Figure 4D *Support Placement*

Ref #	Description
A	Supports at each end of each section

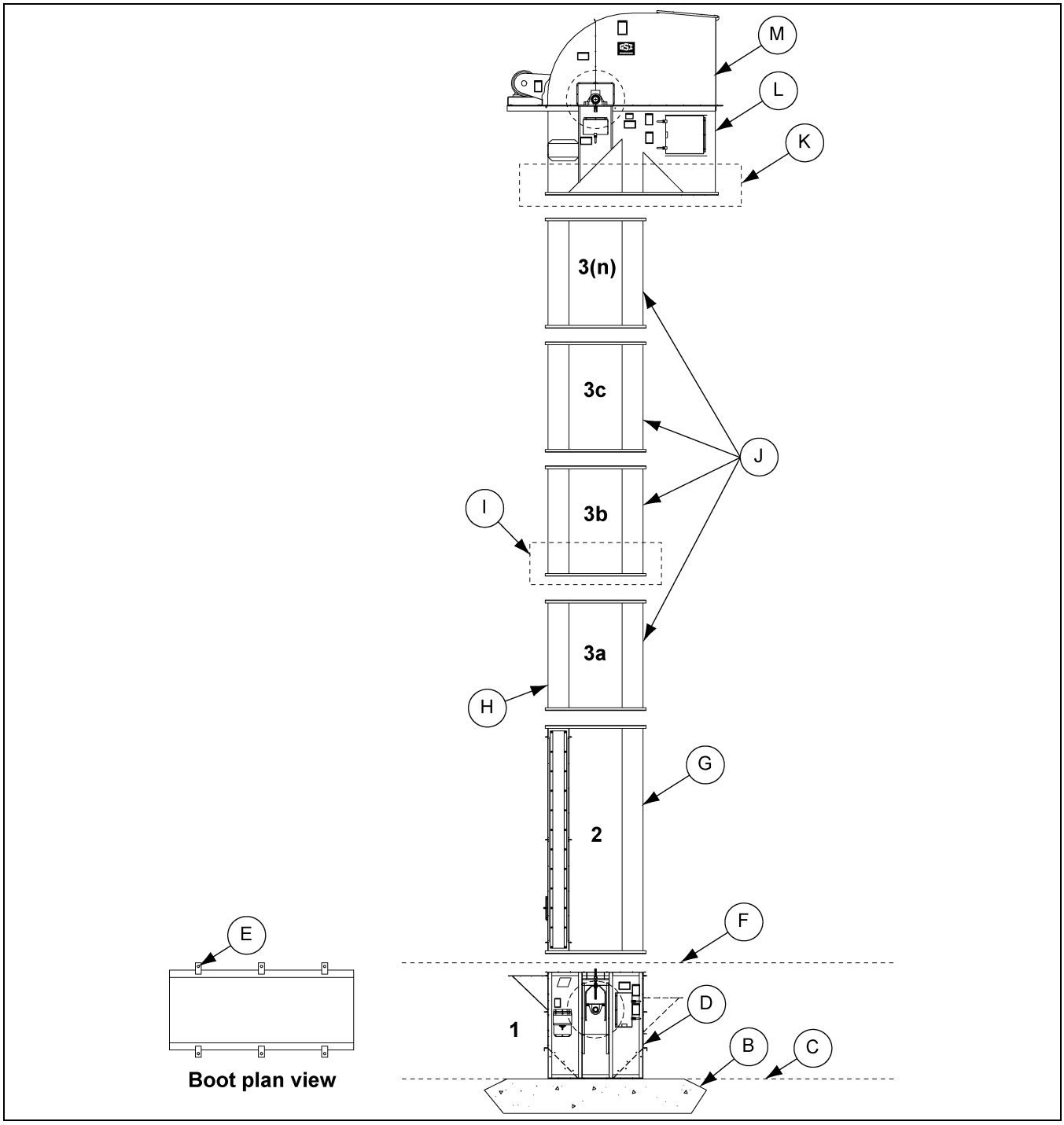


Figure 4E *Vertical Elevator Assembly Sequence*

Ref #	Description
B	Typical Foundation
C	Bottom Level or Grade
D	Boot First Section
E	Expansion Bolt and Anchoring Plates by Others
F	Normal Work Floor Level
G	Typical Inspection Section

Ref #	Description
H	Heavier Gauge Casings if Required
I	Typical Rest Platform
J	3a through 3(n) (Elevator Height)
K	Typical Head Service Platform
L	Head Section
M	Hood Section

Incorrect placement of trunk sections of varying gauge sheet metal could result in support failure, collapse of the elevator and death or severe injury to anyone in the area. Trunk sections of varying gauge must be installed in the locations shown on the certified drawings with the heavier gauges toward the bottom.

4. Begin lifting succeeding elevator sections into position. In most but not all installations, the inspection elevator section is the next section after the boot to be installed.

   a. Certain applications, particularly tall elevators, impose heavy structural loads on the elevator leg and boot sections. In these applications, lower trunk sections and the boot section are fabricated from heavier gauge materials. Packing list indicates quantity of each gauge of trunk. Individual pieces are labeled to show gauge.

   b. Continually refer to the certified drawings, so that trunk sections are correctly positioned for ladder attachment.

   c. If the customer has ordered the elevator with optional explosion vents in certain elevator sections, these sections must be located in their specified locations.

   d. If the elevator includes a short (non-standard) height elevator section, the certified drawings furnished with the elevator will indicate its intended position, usually immediately above the inspection section.

5. Plumb each elevator section as it is positioned on the preceding section. Attach guying and/or bracing as determined by pre-installation planning. Shim as required to plumb each section and tighten assembly fasteners. *(See Figure 4F.)* The elevator shipment included one or more containers of bulk fasteners, sufficient to completely assemble the elevator and any accessories ordered with it. **FASTENERS WERE NOT PROVIDED FOR ASSEMBLY OR ATTACHMENT OF BRACES OR ANY OTHER EQUIPMENT.**

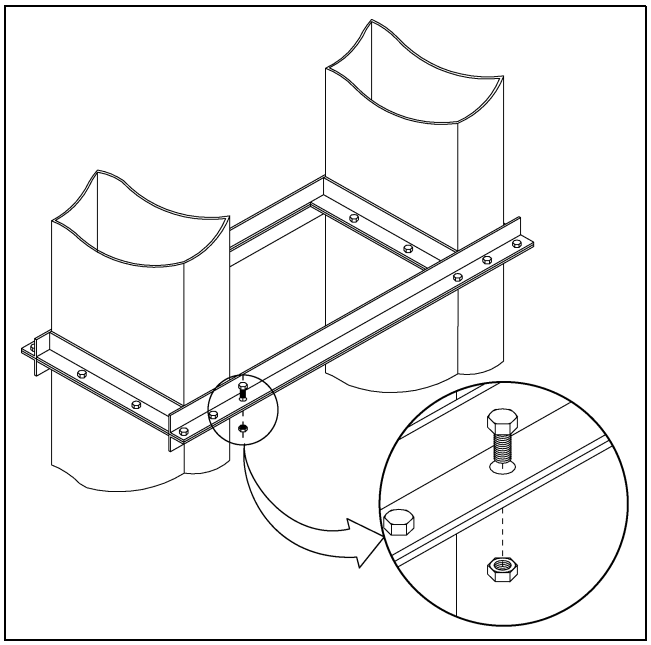


Figure 4F *Elevator Joint Fasteners*

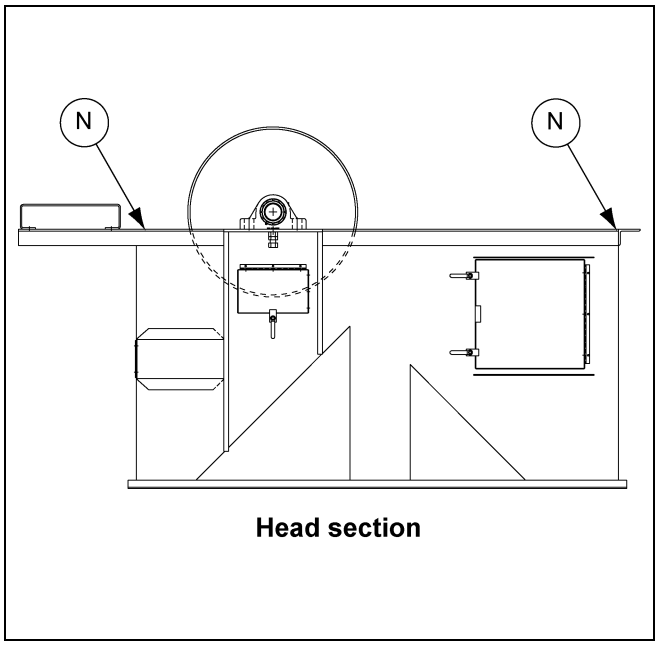
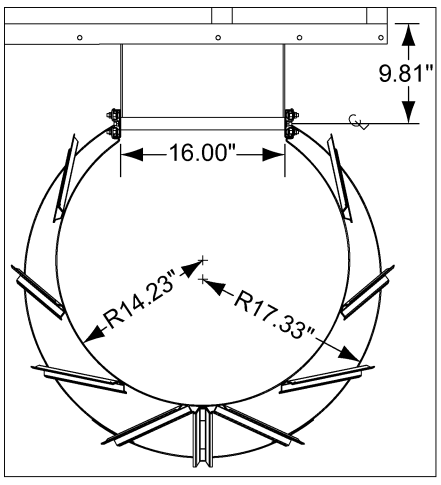


Figure 4G *Lifting Hole Locations*

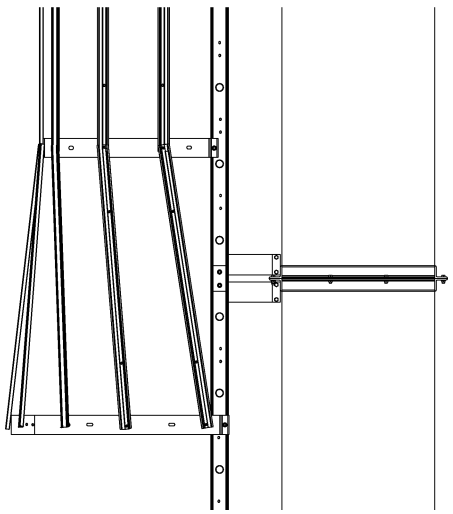
Ref #	Description
N	Lifting Holes

6. The last elevator section to be lifted into position onto the elevator is **ALWAYS** the head section. (The hood is not assembled until belt and cup installation is complete.) Lift the head onto the top elevator section. *(See Figure 4G on Page 23.)* DO NOT install the hood at this time. Plumb the head section and fasten it to the elevator trunk.
7. The basic elevator structure should now be complete, securely braced and/or guyed.
8. Assemble the service platform on the elevator head. If supplied by GSI, a drawing will be furnished, giving specific assembly instructions and a list of specific fasteners for each detail. Do not weld any platform clamping brackets until all platforms and ladders are assembled and correct installation has been verified.
9. Assemble the other service and rest platforms if any were ordered and supplied.
10. Follow these guidelines while assembling ladders:
    a. A length of ladder will lead up to a platform and extend upward through the platform opening to the level of the platform's top rail.
    b. An elevator may include a short section of trunk. In these cases, an extra 10' section of ladder and safety cage is furnished. This ladder and cage section is to be field-modified to match the short section of trunk exactly and compensate for any variances.
    c. The total lengths of safety cage will not equal the total lengths of ladders when rest platforms are used. The bottom edges of the hoops are spaced approximately 7' above the floor level of the platforms.
    d. The bottom of the ladder, whether at ground or grade level or at the working floor, should be surrounded by a lockable safety cage or otherwise arranged to prohibit access to unauthorized personnel.
11. When the installer is confident that **ALL** ladders and platforms are correctly assembled and located, weld all platform support channels to the elevator casing (trunk). The clamp bands used to temporarily position the support channels during installation were **NOT** meant to permanently attach the channels to the trunk.

Ladder Assembly with Safety Cages



Starter cage top view



Starter cage side view

Figure 4H *Typical Ladder and Cage Assembly*

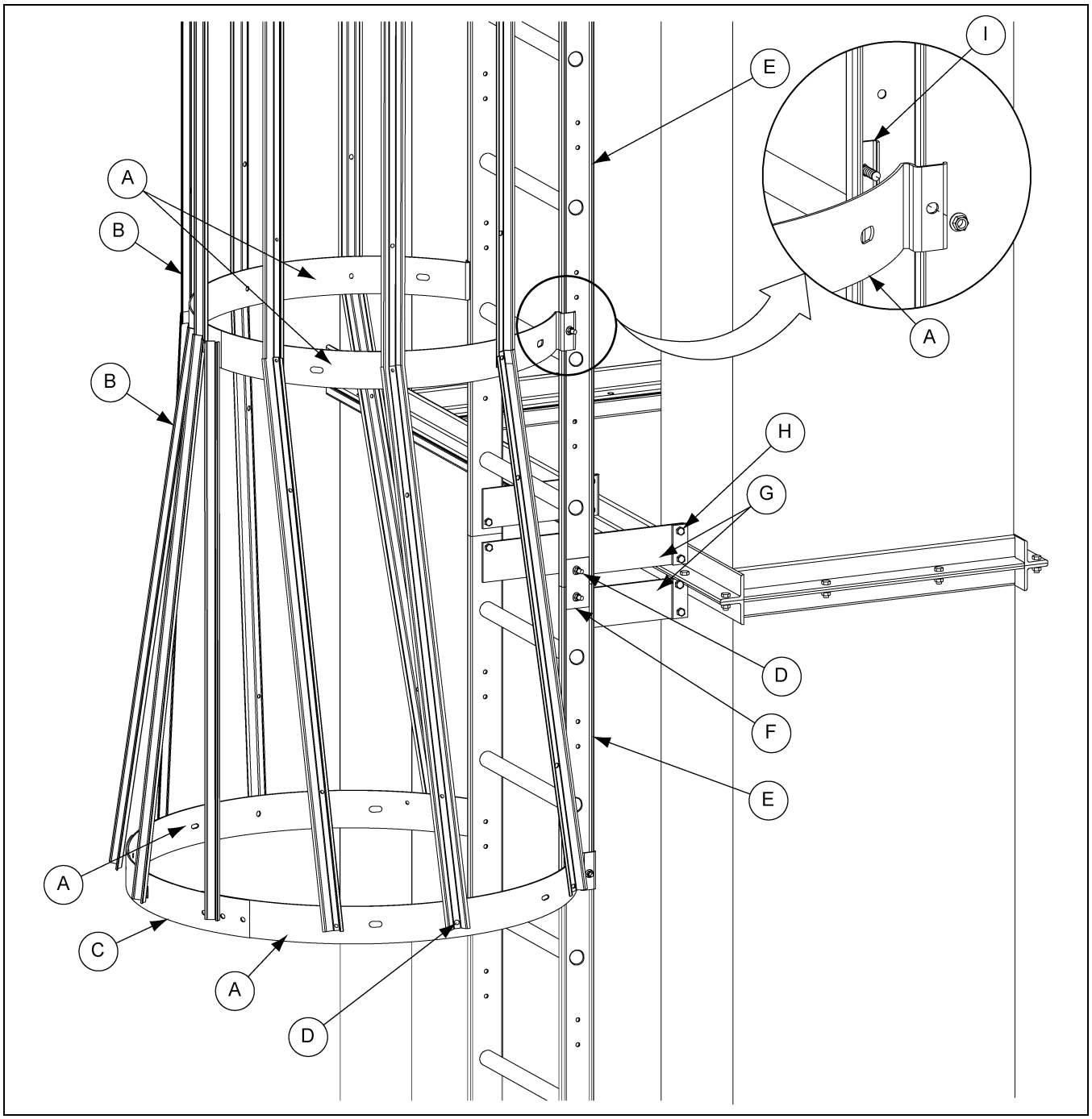


Figure 4I *Standard and Starter Cage*

Ref #	Description
A	Hoop Half
B	Cage Rail Package (44" Sections)
C	Hoop Expander
D	5/16" Hardware
E	Ladder Section

Ref #	Description
F	Ladder Seam Splice
G	Standoff Brackets
H	3/8" Hardware
I	Locking Clip

NOTE: *Repeat the same procees to assemble the remaining safety cages.*

Typical Small Head Service Platform Assembly

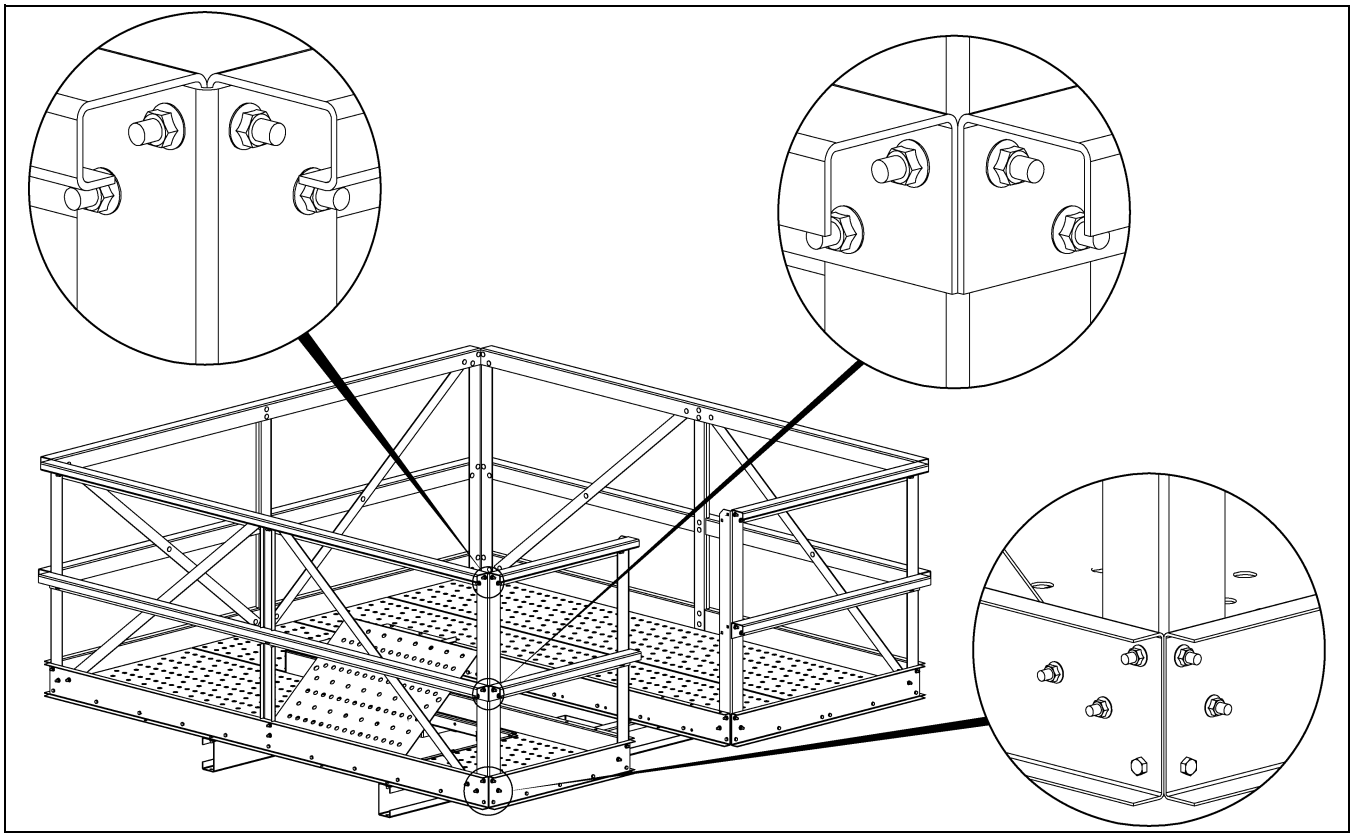


Figure 4J *Typical Small Head Service Platform Assembly*

Typical Large Head Service Platform Assembly

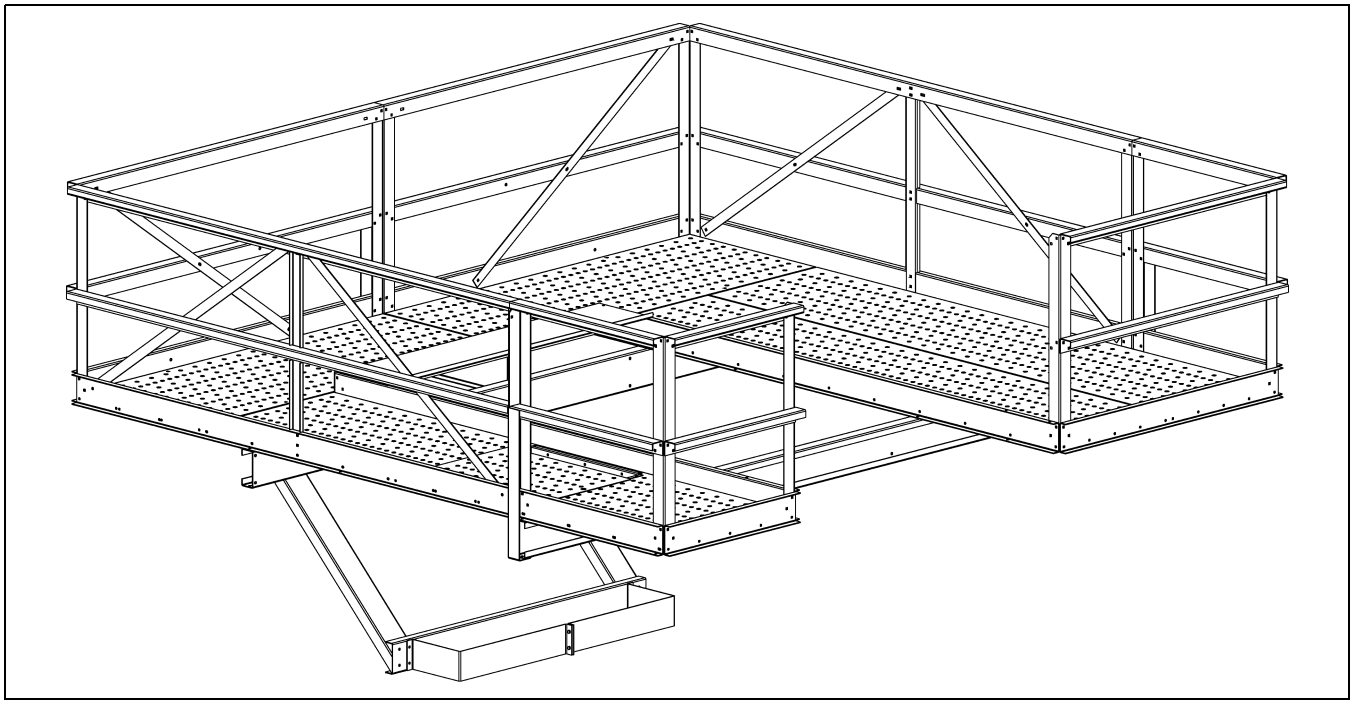


Figure 4K *Typical Large Head Service Platform Assembly*

Typical Head Service Hatch Assembly

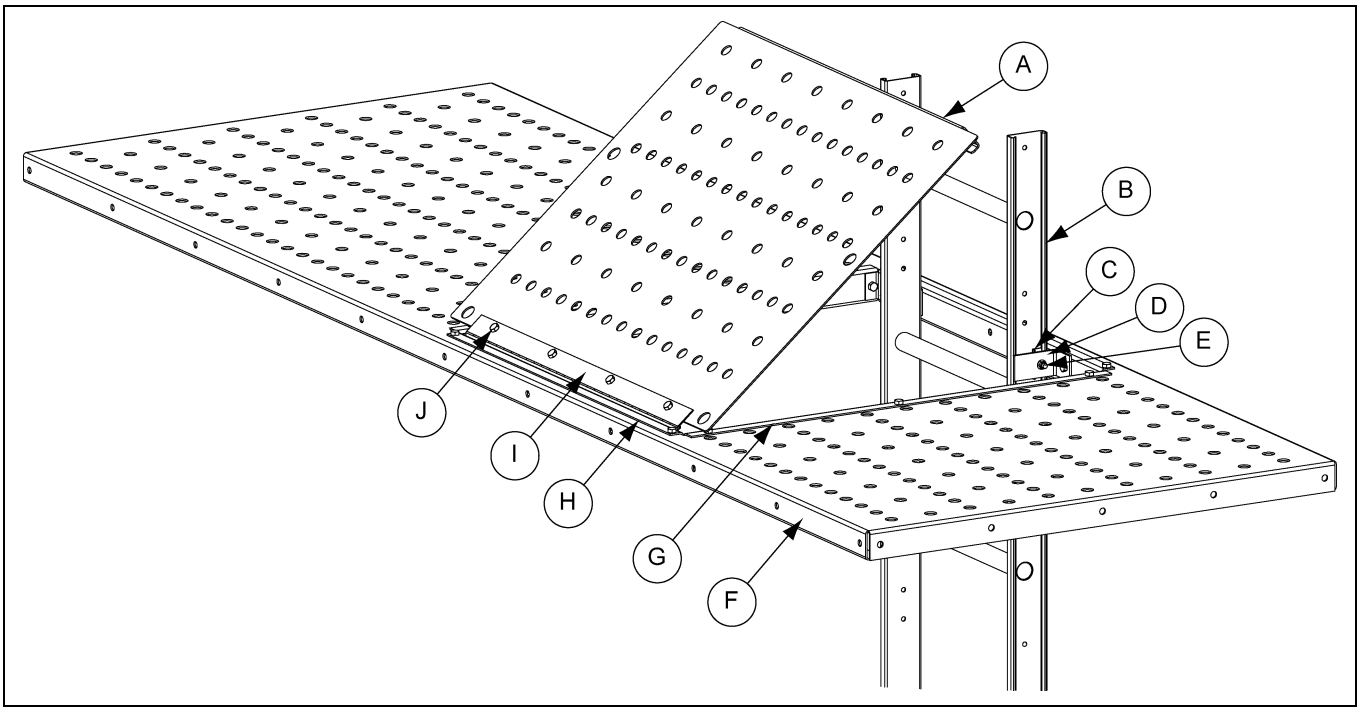


Figure 4L *Typical Head Service Hatch Assembly*

Ref #	Description
A	Hatch Door
B	Ladder Section
C	Locking Clip
D	Ladder Bracket
E	5/16" Nut
F	Perfstrut Deck
G	Hatch Frame Side
H	Hatch Frame End
I	Hatch Hinge
J	3/8" Hardware

Belt and Bucket Installation

There are two (2) basic methods of installing the belt and buckets in an elevator. Any of three (3) styles of belt splice can be selected. Customers usually have a definite preference as to the style of splice used. Various methods of belt/bucket installation and belt splicing are covered in the following paragraphs. *See Figure 4M below* and *Figure 4Q on Page 34*.

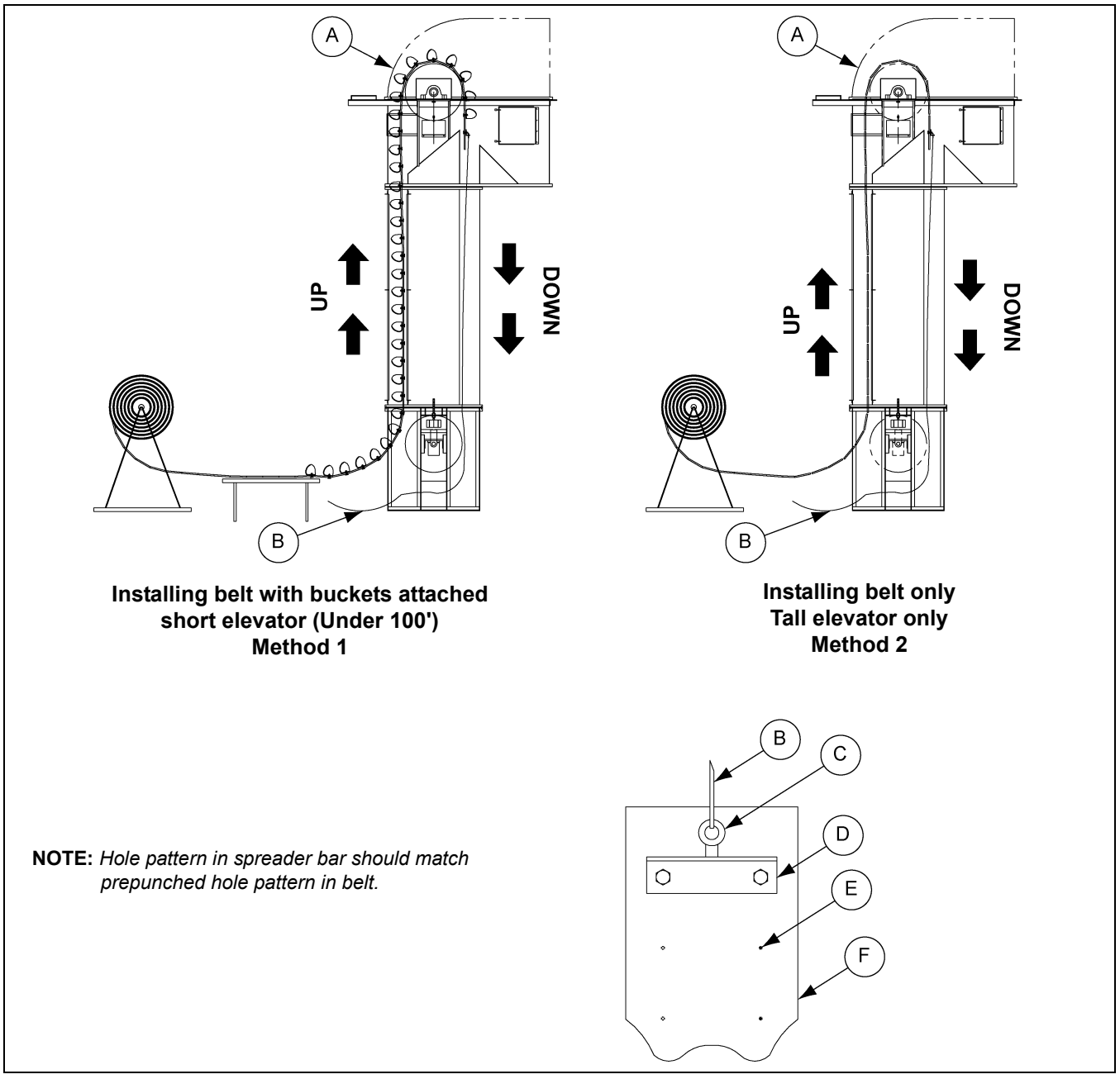


Figure 4M *Belt and Bucket Installation*

Ref #	Description
A	Hood Removed
B	Haulage Line
C	Eye Bolt

Ref #	Description
D	Structural Angle Spreader Bar
E	Prepunched Holes
F	Belt

The choice of methods used to install the belt and buckets is determined by:

1. Length and weight of belt
2. Number, size, and weight of buckets
3. Open area available around the base of the elevator
4. Available equipment

REMINDER: *The hood is not yet assembled to the head section.*

The drawings furnished with the elevator identify the belt to be used, either as furnished by GSI or as specified by the customer, if provided by others. The drawings also specify the type, size, number, spacing, and pattern of buckets on the belt.

1. If the elevator boot includes a manual screw take-up as shown in *Figure 4N*, adjust the take-up screws to raise the pulley to its highest level.

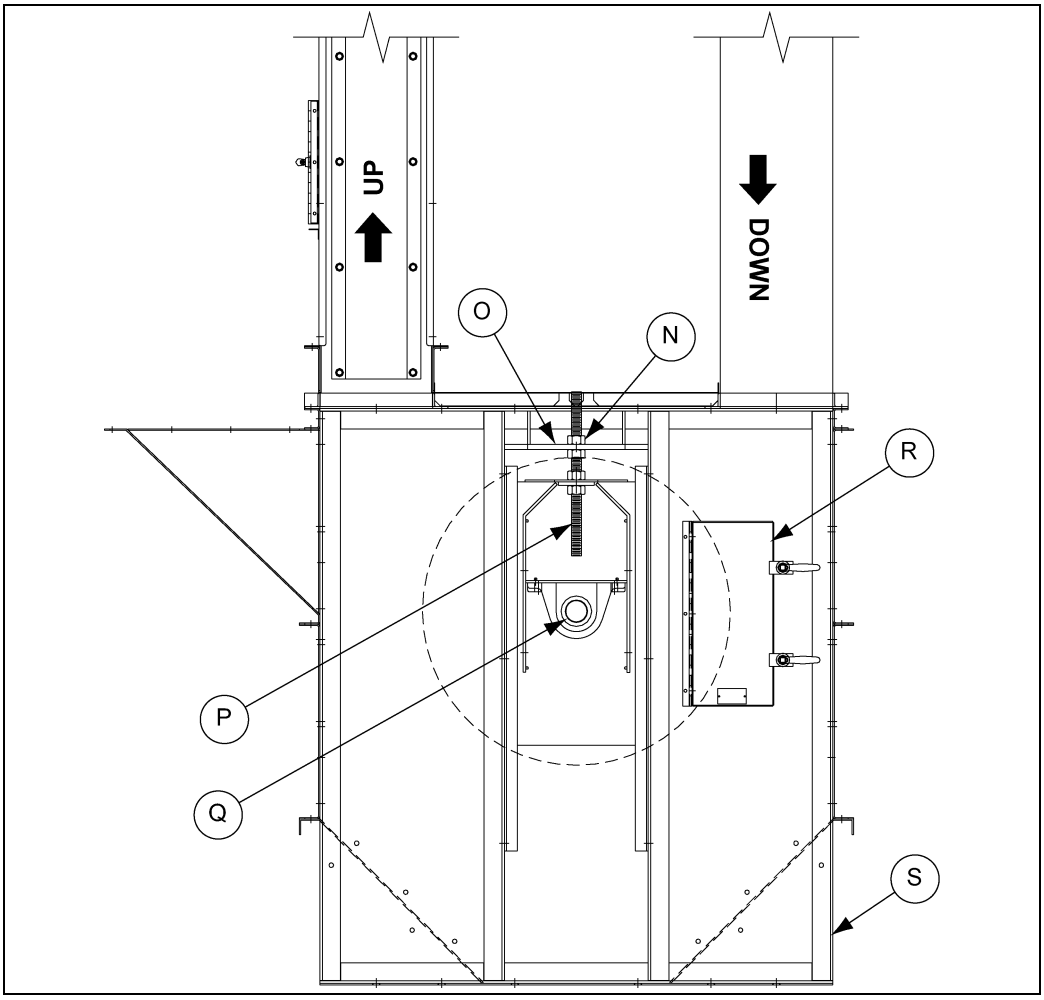


Figure 4N *Screw Take-Up Section*

Ref #	Description	Ref #	Description
N	ACME Nuts	Q	Pillow Block Bearing
O	Support Plate	R	Access Panel
P	Take-Up Screw	S	Boot Assembly

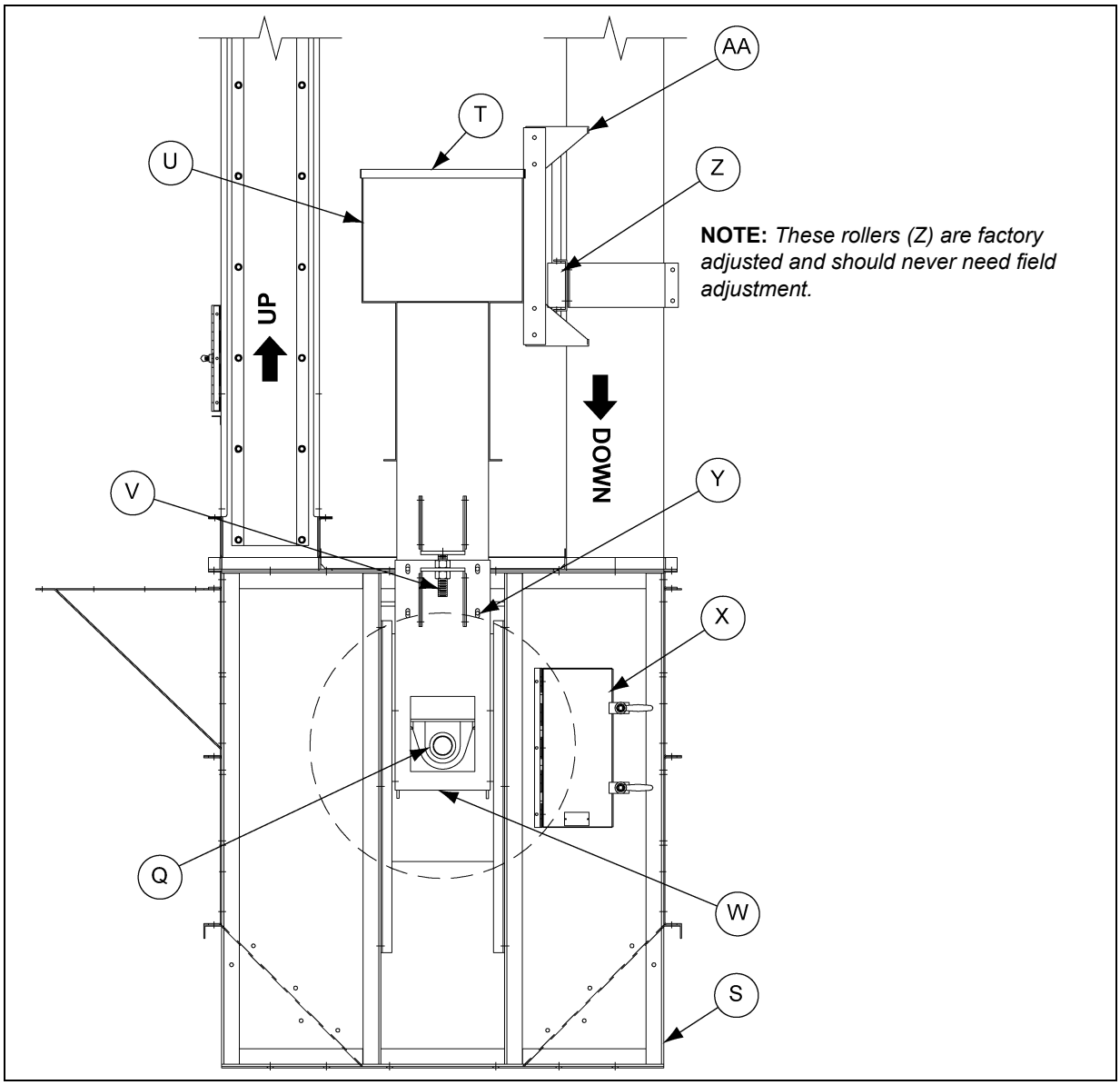


Figure 4O *Posi-Guide Gravity Take-Up*

Ref #	Description
Q	Pillow Block Bearing
S	Boot Assembly
T	Removable Cover
U	Weight Box
V	Alignment Adjusting Screws

Ref #	Description
W	Lift Here to Raise Pulley
X	Access Door
Y	Connection Bolts
Z	Guide Rollers
AA	Roller Bracket

2. If the elevator includes a posi-guide gravity take-up as shown in *Figure 4O*, position a jack under the indicated surface of the take-up side post to lift the take-up, including the pulley.

WARNING ***An improperly supported posi-guide gravity take-up can slip out of position or fall, causing serious injury or damage. Assure proper placement of jack and supports.***

3. Rig a heavy haulage line (rope or cable) through the removable panel in the UP-LEG through which the belt with buckets attached will be pulled into the elevator. The line is then hauled up the **UP-LEG**, over the head pulley, and down the **DOWN** elevator leg. Usually, the line can be passed around the boot take-up pulley and out of a cleanout panel to a winch so the belt can be pulled into the elevator. A spreader bar should be fabricated to connect the haulage line to the belt. See *View D* of *Figure 4Q on Page 34*. The bar is preferred to simply punching a hole in the belt and tying or clamping the line to the belt.
4. "ACME" Nuts should both be tightened against support plate to prevent vibration from moving the take-up out of alignment. Should the take-up screw ever require replacement make certain to loosen both "ACME" Nuts to relieve any upward or downward tension before removing the retainer bolts.
5. The connection bolts must be loosened while aligning the pulley with the adjusting screws and re-tightened when alignment is complete. The weight of the take-up, boot bearings, etc., provides proper belt tension in most applications. For start-up, begin with the gravity take-up weight box empty. If the belt slip occurs, stop the elevator and add ballast in 100 lbs. increments until slipping stops. Evenly distribute the weight in the weight box. Do not exceed the maximum allowable ballast weight shown on the elevator drawing. Excessive weight will cause belt damage.

Method 1: Simultaneous Belt and Bucket Installation (Less than 100')

IMPORTANT: *Omit at least three (3) rows or sets of buckets at each end of the belt until after the belt has been spliced.*

1. Place a flat table in-line with an opening in the Up-leg trunk of the elevator, usually the end panel in the boot section. Remove the boot end panel.
2. Arrange the roll of belt so it can easily be unrolled onto the worktable and then into the elevator's UP-leg. Follow the belt manufacturer's recommendations, if any, as to which side of the belt should contact the pulleys and which side should face outward and contact the buckets.

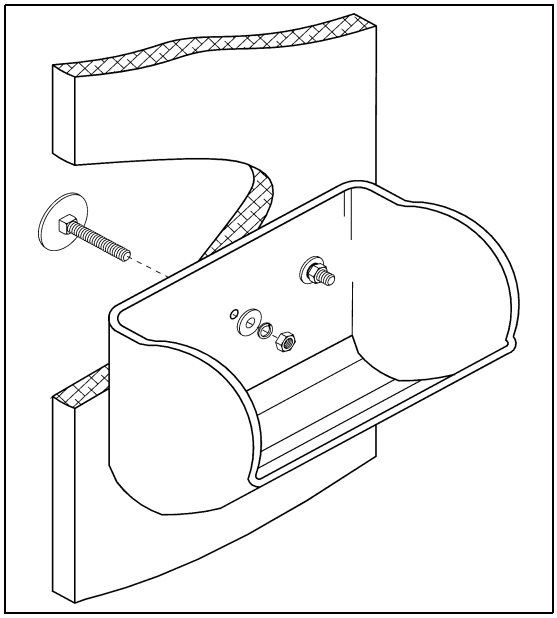


Figure 4P *General Details of Bucket Attachment*

3. The belt should already have the bucket mounting holes prepunched in the appropriate pattern. Begin bolting buckets to the belt in the pattern and at the spacing shown on the drawing. The heads of the elevator bolts bear against the inside surface of the belt. Buckets are to bear against the opposite or outer surface of the belt. Elevator bolts should be tightened so as to draw the heads flush with the belt surface. This prevents the bolt heads from scarring the pulleys or working loose. Refer to hardware manufacturer for recommended installation and torque criteria.

4. Connect the haulage line to the end of the belt, preferably by means of a spreader bar.

5. As buckets are fastened to each length of belt, pull the length of belt into the elevator leg. Be sure to keep a strain on the haulage line between hauls so the belt does not drop back down the elevator. As more belt is hauled into the elevator and more strain is imposed on the haulage line, make certain that the line cannot slip and is of proper strength.

   IMPORTANT: *Before hauling the first length of belt with buckets attached, check again to make certain that the buckets are being attached with the bucket openings facing upward in the "up" leg of trunk.*

6. As the belt begins to pass over the head pulley, keep the belt roughly centered. Proper belt tracking will be done when belt installation is complete.

7. When all buckets except those at the ends of the belt where the splice is to be made have been fastened to the belt AND the entire belt has been hauled into the elevator, splice the belt per belt splicing *on Page 33*.

8. Fasten any remaining loose buckets to the belt; in the area of the splice, for instance.

9. Proceed with initial belt tracking. Refer to initial belt tracking *on Page 36*.

Method 2: Belt Only With Buckets Attached after Splicing Belt

1. Arrange the roll of belt so that it can easily be unrolled and hauled into the elevator's UP-LEG. Follow the belt manufacturer's recommendations, if any, as to which side of the belt should contact the pulleys and which side should face outward and contact the buckets.

2. Connect the haulage line to the end of the belt, preferably by means of a spreader bar.

3. Begin hauling the belt into the elevator. As more belt is hauled into the elevator, more strain will be imposed on the haulage line. Keep a sharp eye on the line for any sign that it might part or slip on the winch.

4. As the belt passes over the head pulley, keep the belt roughly centered. Proper belt tracking will be done when belt installation is complete.

5. Splice the belt. Attach the buckets to the belt in the splice area NOW so the fasteners do not have to be loosened later. Refer to belt splicing *on Page 33*.

6. Adjust the boot take-up pulley to apply tension to the belt. Refer to *Step A* in initial belt tracking *on Page 36*.

7. It will be necessary to use the elevator drive or some other means to rotate the belt to allow the buckets to be fastened to it.

   IMPORTANT: *If the elevator drive is operated to rotate the belt, make sure to lubricate the reducer and all other components which require initial lubrication before applying power. Damage will result to any component operated without proper lubrication.*

8. Fasten a bucket (or a set of buckets, if there are several buckets in each horizontal row in the pattern shown on the drawings) to an accessible set of mounting holes.

9. Skip five (5) or ten (10) hole patterns. Fasten another bucket, or set of buckets, to the belt. Continue this sequence until the belt has made one complete revolution through the elevator. Then start fastening buckets or sets of buckets in the center of the gap between the previously installed buckets.

10. This procedure uses buckets as counterweights. If buckets are fastened to the belt in consecutive continuous rows, the strand of belt in the **UP-LEG** will carry appreciably more weight than the strand in the **DOWN-LEG**. Considerable backward pull on the head pulley would result and it would be much more difficult to advance the belt. Continue the sequence until the full complement of buckets has been fastened to the belt.

11. When bucket installation is complete, complete initial belt tracking as shown *on Page 36*.

Belt Splicing

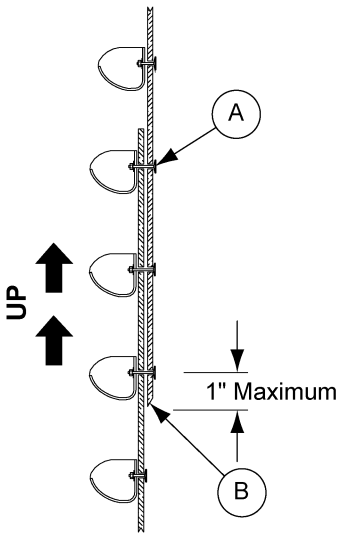
There are three (3) types of belt splices normally used with elevator belts and each is described in the following paragraphs. The splicing method should have been selected during installation preplanning as explained in Elevator Belt Selection *on Page 20*.

IMPORTANT: *Regardless of the belt splicing method selected, it is vitally important that the splice be perfectly aligned. The belt edges at the splice must be absolutely parallel to permit tracking of the belt and satisfactory elevator operation.*

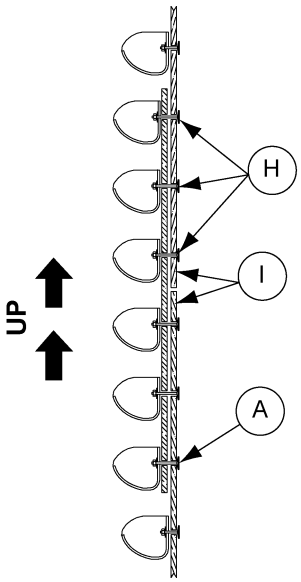
1. Lap Splice: *(View A on Page 34.)* This is the easiest splicing method. Make sure the belt ends overlap as shown so the inner belt end does not snag on the pulleys, especially the head pulley which is often lagged or winged.

2. Butt Splice: *(View B on Page 34.)* This method of belt splicing makes for a smoother running belt as there is no ridge at the splice as with the lap splice. Somewhat more belt is required and resplicing takes somewhat more time than a lap splice since twice the number of buckets must be removed and reinstalled. On the plus side however, the resultant splice is stronger.

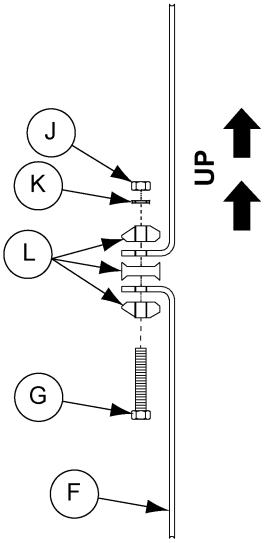
3. Mechanical Splice: *(View C on Page 34.)* This is a superior splice. It makes for the smoothest running belt. However, it is also the most difficult splice to make properly. Vulcanized and cemented splices do not often find application in elevators. The splices are difficult to apply in the field. Also, the belt may have to be respliced after a short period of operation to compensate for initial stretch. Regardless of the method used to splice the belt, a set of spreader bars and a cable hoist as shown in *View D on Page 34* will facilitate splicing.



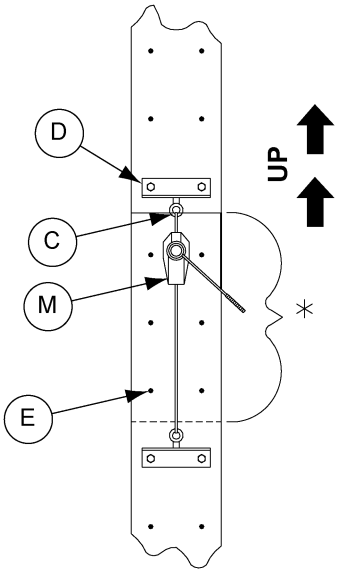
View A: Lap Splice



View B: Butt Splice



View C: Mechanical Splice



Three (3) bucket splice example:

- For a Lap splice, overlap 3 bucket spaces.
- For a Butt splice, overlap 6 bucket spaces.

View D: Spreader Bar Detail

Figure 4Q *Belt Splicing Methods*

Ref #	Description
A	Longer bolts are used to attach buckets where belt is spliced.
B	Bevel Pulleyside End
C	Eye Bolt
D	Structural Angle Spreader Bar
E	Prepunched Holes
F	Belt
G	Bolt

Ref #	Description
H	Three (3) bucket overlap on each side of seam between splice ends.
I	Ends of Main Belt
J	Nut
K	Lock Washer
L	Splice Clamps
M	Cable Hoist

Hood Installation

After belt splicing is complete, hoist the hood into position on the elevator head and install the fasteners.

Drive Installation

Refer to the order information and the certified drawings furnished with the elevator for drive installation. The amount of detail will depend upon whether or not GSI furnished all or part of the elevator drive. Abide by the manufacturer's instructions packaged with the reducer, motor, couplings, bushings, etc. Fit and alignment must be held to the limits specified to achieve the expected service life. In any event, the engineer or contractor overseeing elevator installation is responsible for proper installation and operation of the elevator drive.

IMPORTANT: *Reverse rotation of drive can cause damage to internal elevator and/or drive components. Be certain to verify correct motor rotation on start-up. If motor rotates in the wrong direction, change motor wiring to achieve proper rotation.*

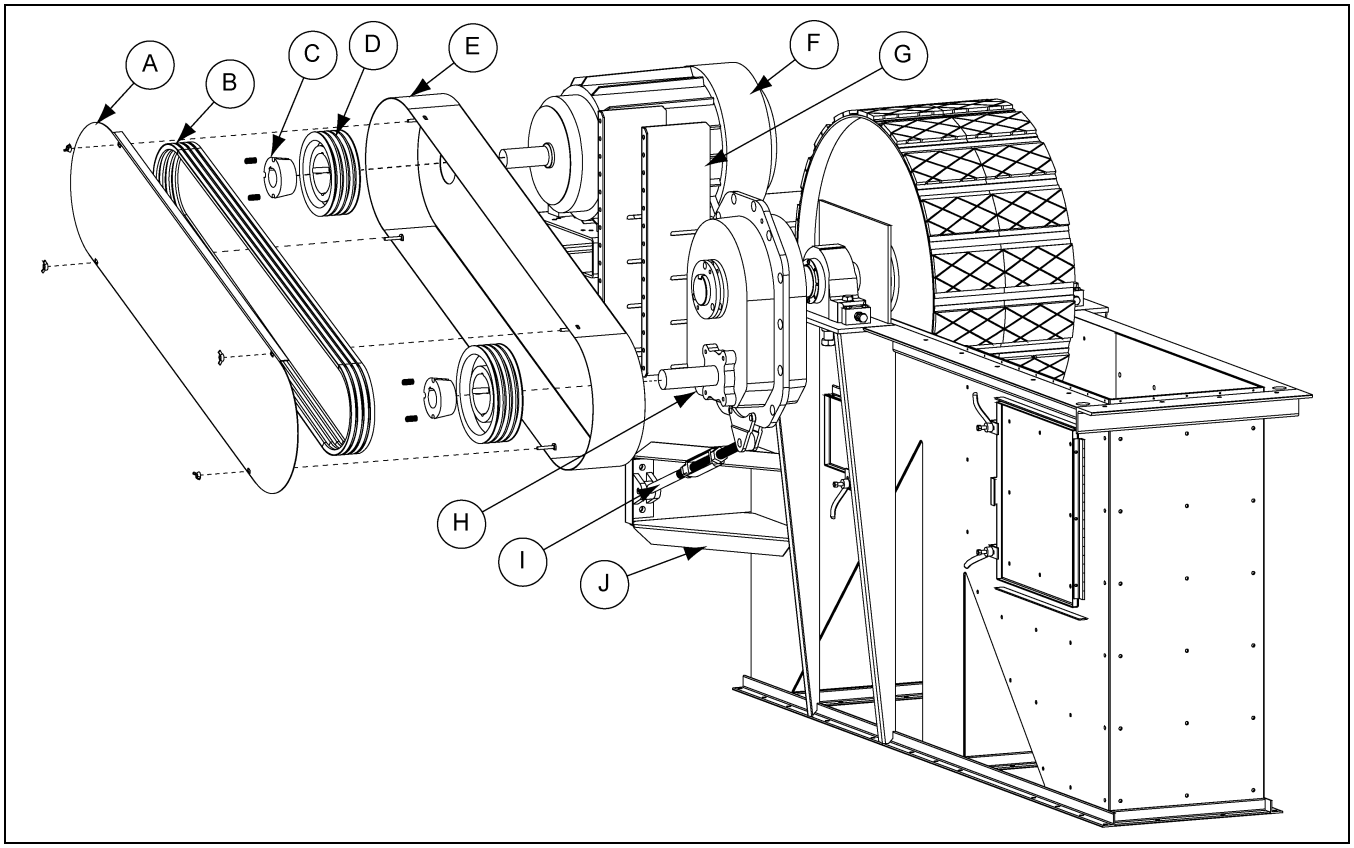


Figure 4R *Typical Shaft Mount Drive Assembly*

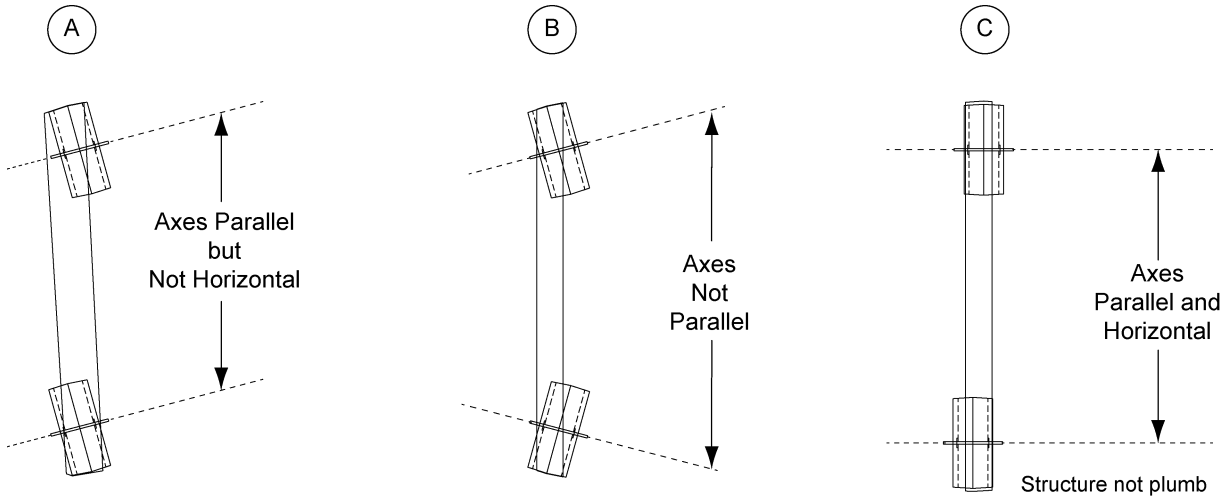
Ref #	Description
A	Belt Guard Cover
B	Belts
C	Bushing
D	Sheave
E	Belt Guard Body

Ref #	Description
F	Motor
G	Belt Guard Mounting Bracket
H	Shaft Mount Gear Reducer
I	Gear Reducer Torque Arm
J	Torque Arm Gusset

Initial Belt Tracking

IMPORTANT: *This procedure is to be performed with the elevator empty. Should any problem be discovered, it will be much more difficult to correct if the elevator is full of material.*

1. When the belt has been spliced, shift the belt laterally on the head pulley so it is centered on the pulley.
   a. If the elevator has a screw take-up, *Figure 4N on Page 29*, adjust the screws evenly downward at both ends of the take-up pulley shaft. Continue adjusting the screws until the belt can be seen to wrap snugly around the take-up pulley.
   b. If the elevator has a posi-guide take-up, *Figure 4O on Page 30*, then simply remove the temporary support so the weight of the take-up is exerted on the take-up pulley. If the loaded elevator, when tested, is found to need more belt tension, that can be accomplished by adding ballast to the weight box.
2. Establish a firm, level, horizontal reference plane common to both ends of the take-up or boot pulley shaft. Measure from the reference plane to the pillow block mounting surface at each end of the pulley shaft. If the distances are not equal, adjust the take-up screws, *Figure 4N on Page 29*, or the alignment adjusting screws, *Figure 4O on Page 30* until the pulley shaft is level.
3. Establish a firm, level, horizontal reference plane common to both ends of the head pulley shaft. Measure from the reference plane to the pillow block mounting surface at each end of the pulley shaft. See *Figure 5B on Page 38.*
   a. If both mounting surfaces and hence both pillow blocks are not at the same height, loosen the mounting bolts for the pillow block on the low end of the head shaft.
   b. Turn the jack screws evenly to raise the pillow block until both ends of the head shaft are level. Shim between the pillow block and its mounting surface. Shims should be as nearly equal to the area of the pillow block base as possible so as not to concentrate stress on a small area of the bearing housing , causing it to fracture. Slotted shims are suitable and easy to use.
   c. After the pillow block has been shimmed to level the head pulley, back off the jack screws so the full weight of the pillow blocks rests on the shims. Tighten the mounting screws.
4. If it has not already been done, lubricate all components requiring initial lubrication. This includes, but is not limited to, the drive reducer. Reducers are usually shipped dry, without lubricant. Refer to the manufacturer data, which should have been saved when the components were unpacked or consult the manufacturer or a local supplier of the component.
5. Start the elevator. Observe the belt as it passes around the pulleys. If the belt appears to be “walking” towards the edge of the pulley, be ready to shut down the elevator drive. Several complete revolutions of the belt will be required before the belt reaches a steady-state tracking condition.
6. If the belt steadily tracks on or near the center of both pulleys, no further action is necessary. If, however, the problem is that the elevator structure is significantly out of plumb, the remedy is going to be more complex. The elevator will have to be plumbed **NOW** after all the guying and/or bracing has been installed. Study the diagrams in *Figure 5A on Page 37*. Try to correct belt tracking first by adjusting the boot take-up pulley and then by shimming the head pulley pillow block bearings. If altering pulley alignment has no effect, run the elevator until the belt splice is accessible through the inspection door. Use a long straightedge (10' or longer) or taut line to see if the belt was spliced squarely. If necessary, resplice the belt.

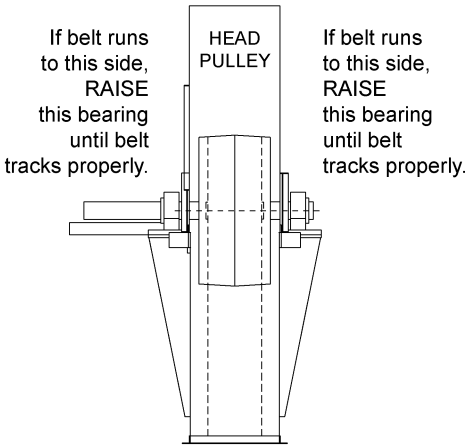


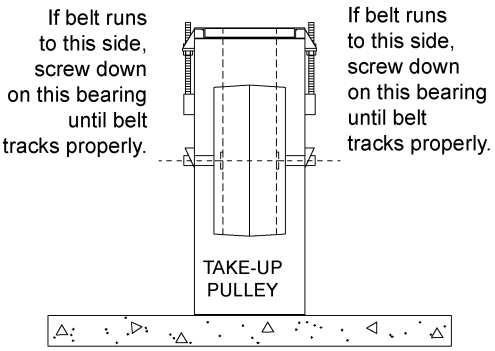
Shim head pulley bearing and adjust take-up pulley

Shim head pulley bearing and adjust take-up pulley

Realignment of elevator section necessary

* Rubbing pulleys are a serious fire hazard and should be corrected IMMEDIATELY.
* Minor adjustments may be made as shown in the illustrations.
* Serious defects should be corrected by a qualified millwright.





Other Causes of Head Pulley Misalignment

1. Loose set screws or keys, allowing the pulley to move laterally on the shaft.
2. Loose bearing or bearing support.
3. Weak or bent head pulley shaft.

Other Causes of Boot Pulley Misalignment

1. Grain entering the boot at an angle crowding the belt to one side.
2. Loose set screws or keys, allowing the pulley to move laterally on the shaft.
3. Head pulley out of plumb.

Figure 5A *Belt Tracking Guidelines*

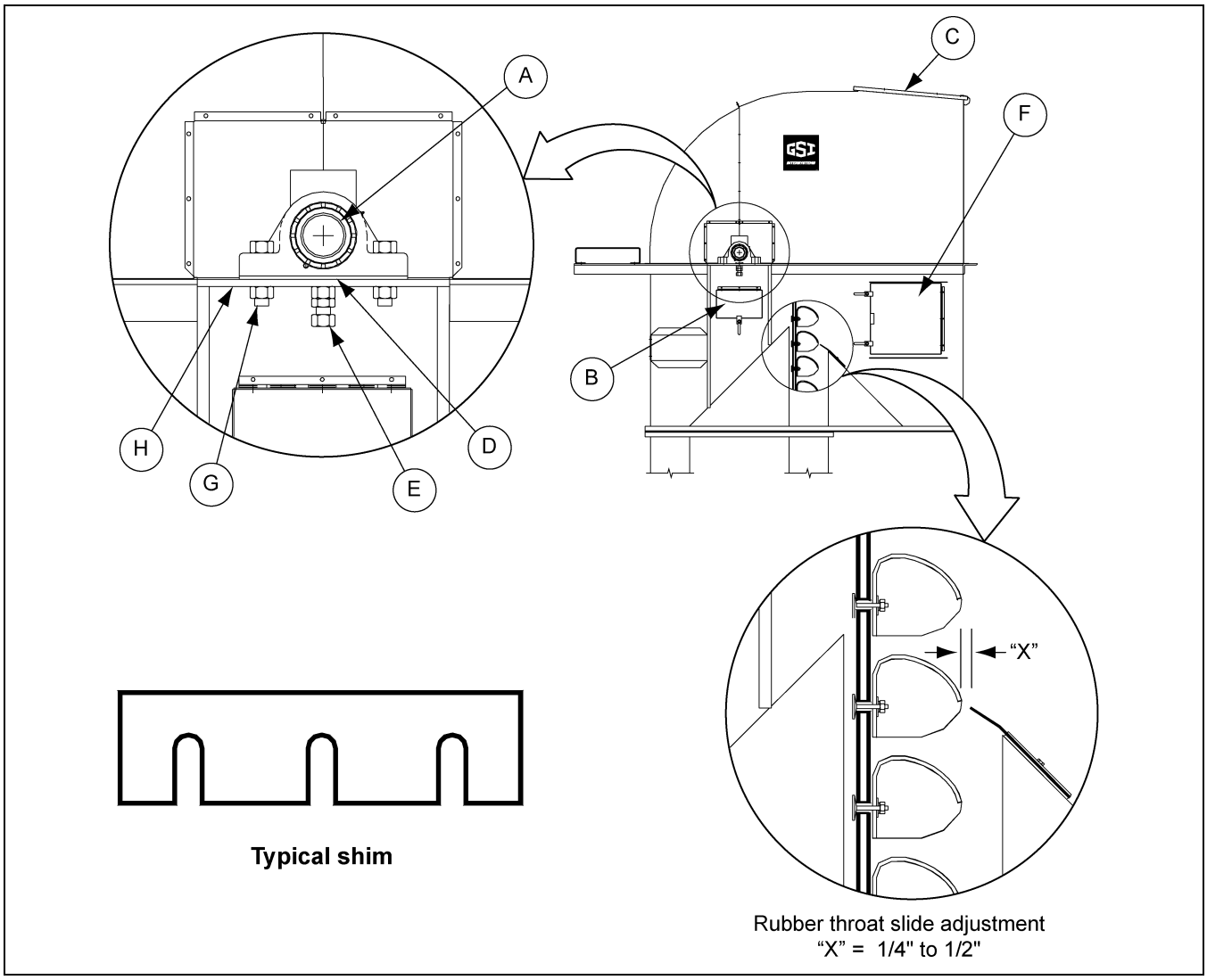


Figure 5B *Head Section Details*

Ref #	Description
A	Head Pulley Shaft
B	Slide Lag Inspection and Replacement Panel - Hinged
C	Explosion Vent
D	Shim

Ref #	Description
E	Jack Screw
F	Rubber Throat Slide Access Panel - Hinged
G	Mounting Hardware
H	Pillow Block Mounting Surface

Rubber Throat Slide Adjustment

The rubber throat slide, as shown in *Figure 5B*, should be positioned so there is 1/4" to 1/2" clearance between the edge of the slide and the outer lips of the buckets. Proper adjustment will minimize one cause of down-legging.

IMPORTANT: *For lap or butt spliced belt, increase the clearance dimension between the rubber throat slide and the buckets by the thickness of the belt to compensate for the double belt thickness.*

In-Service Belt Tension

If the belt tracks properly and does not slip when the elevator is operating at maximum capacity, then the belt is sufficiently tensioned. Tension adjustment, however, must wait until the entire installation is complete and material (the product; grain, etc.) can be conveyed to and from the elevator.

1. While observing all prudent safety precautions, posted or otherwise, examine the belt as it passes over the head pulley **WHILE** the elevator is under load. The pulley should track properly. Belt slip is most often detected by observing the bearing setscrews on the head and tail shafts while counting the number of revolutions of pulley in the same period of time, typically 30 or 60 seconds.

IMPORTANT: *Since the head pulley is lagged and its effective diameter is larger compared to the boot pulley, head pulley rotation speed will be slightly slower than the boot pulley even if there is no belt slip.*

If the belt is slipping and all other conditions appear to be normal, adjust the take-up.

a. If the elevator has a screw take-up, adjust the acme screws evenly to force the pulley downward and increase belt tension.

(OR)

b. If the elevator has a posi-guide gravity take-up, adding ballast to the weight box will increase belt tension. The ballast should be spread evenly throughout the box for best results. Be sure to reinstall the weight box cover to prevent rainwater from accumulating in the weight box. See *Figure 4O on Page 30*.

2. Verify that belt speed closely approximates that specified on the drawings. If it does not, check the drive system, especially if all drive components were not furnished by GSI. Among the points to consider are:

   a. If the elevator includes a chain drive, compare the number of teeth of the drive and driven sprocket. Verify that the resultant ratio, taken together with the reducer's output speed, will produce the specified final pulley RPM.
   If the elevator includes a V-belt drive, compare the pitch diameters of the drive and driven sheaves. Verify that the resultant ratio taken together with the reducer's output speed will produce the specified final pulley RPM.

   b. Whether the elevator employs direct coupled, chain, or belt drive, you must verify that the motor's speed and the reducer's reduction ratio will produce the required output speed to produce the specified final pulley RPM.

   c. If no other problem is uncovered, check to see that the drive components are not slipping on the shafts. Sometimes during field assembly, the keys are inadvertently left out.

3. If the belt speed matches the specified value and the elevator does not seem to be handling the specified capacity, review the factors set forth in elevator intake *on Page 19* and in elevator discharge *on Page 20*. The elevator will not operate at maximum capacity if material does not flow freely into and out of the elevator.

4. Verify that the belt continues to track the center of the head and boot take-up pulleys. If it does not, check to see if the material being fed into the elevator is entering at an angle from one side or the other and forcing the belt to one side of the pulley. This condition, if it occurs, must be corrected. Continual lateral pressure on the belt can cause the belt edge to rub against the elevator structure. At the very least, the belt will wear rapidly. More serious than wear, a fire and/or explosion can occur if belt friction generates sufficient heat.

5. After the elevator has operated for a period of time, the take-up may require readjustment to compensate for mechanical belt stretch. Depending upon the amount of mechanical stretch, the belt may even have to be shortened and respliced.

Elevator Bucket Filling

Proper filing of the buckets is vital to achieving the design capacity of the elevator. Buckets should be filled to slightly above the water level line shown in *Figure 5C*. In section elevator intake *on Page 19*, discusses some of the factors influencing elevator capacity in general and bucket filling in particular.

The following procedure can be used to give a reasonably accurate capacity indication for an elevator leg when used without a bulkweigh system. For calibration with the use of a bulkweigh system, refer to *Page 41.*

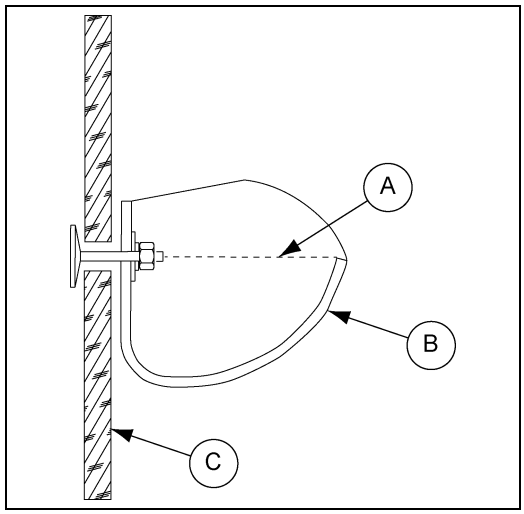


Figure 5C *Bucket Water Line Drawing*

Ref #	Description
A	Water Level
B	Cup
C	Belt

1. Make certain that the pit, all spouts, and the legs are all empty. Be certain you know the exact empty and gross weight of the truck, train car or whatever you are using for the test.
2. Start the elevator and make certain it is running properly and ready to receive material. Load the pit but do not start the feed to the elevator.
3. Open the feed to the elevator and start the timer.
4. It is very important to keep the pit full in order to keep a constant feed rate to the elevator. Continue loading the pit until the truck, train, etc. is empty. When the pit empties, stop the timer.
5. After the entire system empties, use the formula below to calculate BPH capacity.

   Net material weight = (Gross weight truck) - (empty weight)

   TF = 60 / (minutes to run the load)

   (net material weight / (lbs. per bushel of material) x (tf) = BPH

   (OR)

   (net material weight) / (lbs. per cu ft of material) x (tf) = cu FPB

6. Repeat procedure if necessary to confirm results.

EXAMPLE:

Gross truck weight with corn (56# PER BU)	35,000
Empty truck weight	10,000
Time from opening pit gate to pit empty	5 min

35,000 - 10,000 = 25,000 net pounds of corn

60 / 5 min = 12 (tf)

25,000 lbs. / 56 lbs per bu. x 12 (tf) = 5357 BPH

The following procedure can be used to give a reasonably accurate capacity indication for an elevator leg when used **with a bulkweigh system**. For calibration without the use of a bulkweigh system, see the previous page.

1. Make certain that the pit, all spouts, the leg and the bulkweigh system are all empty.
2. Start the elevator leg and make certain it is running properly and prepare bulkweigh system to receive material. Load the pit but do not start the feed to the elevator.
3. Open the feed to the leg and start the timer.
4. After exactly 3 minutes, shut down the feed to the leg.
5. After the entire system empties, check the throughput on the bulkweigh system and use the formula below to calculate BPH capacity.

   1 lbs. through bulkweigh / lbs per bushel of material x 20 = BPH

   OR

   1 lbs through bulkweigh / lbs per cu ft of material x 20 = cu FPH

6. Repeat procedure, if necessary, to confirm results.

   EXAMPLE:

   Handling wheat @ 60 lbs per Bu.

   After 3 min run, 35,000 lbs went through bulkweigh system

   35,000 / 60 x 20 = 11,667 BPH

If Unable to Get Capacity

1. Check for obstruction in inlet spouts or boot hoppers.
2. Check slope of inlet spouts and hoppers. Should be 45° or greater.
3. Check height of boot inlet hopper, where grain entered boot, both up-leg and down-leg.
4. Check belt tension--adjust boot pulley take-ups.
5. Check position of pulley. Take-Ups may be carrying pulley too high for cups to dip into grain.
6. Varnished or painted cups must be given time to polish up for free grain discharge.
7. Check cups for caking or deformation.
8. Check head pulley RPM.

IMPORTANT: *Thoroughly read safety information from Pages 5-16 before beginning installation and start-up.*

General Maintenance

A good maintenance program involves thorough general housekeeping, periodic inspection, adequate lubrication, and timely adjustment.

NOTE: *Never service any equipment while operating unless it is necessary; for example, bearing lubrication.*

General Housekeeping and Periodic Inspection

At frequent and regular intervals, perform these housekeeping chores and inspections:

1. Remove accumulated dirt from the motor, reducer housings and bearings.
   a. Motors depend upon unobstructed airflow over their housings for effective cooling.
   b. Reducer gear cases must also be free of dirt for effective heat radiation. Most reducers have a pressure vent to permit escape of vapors, which may build up internally. If dirt blocks a vent, internal pressure can rupture seals. Leaking lubricant will contaminate the product being handled by the elevator and reducer failure and subsequent equipment downtime will definitely result if the leak is not discovered in time and repairs made. Original equipment manufacturers will refuse to honor warranties if a motor or reducer fails because dirt accumulation leads to overheating or lubricant failure.
   c. While cleaning the reducer, check the reducer's lubricant level and condition. If the level is low, find and correct the leak. If the lubricant is dirty or shows signs of overheating, schedule a change of lubricant as soon as possible.
   d. Listen carefully for a noisy motor, reducer, or bearings, or a rubbing belt. Any of these sounds can be a forewarning of overheating and fire or explosion. Correct any problem discovered immediately.
2. Periodically remove the drop-down clean out door from each end of the boot. Cleanse the boot of all accumulated dirt and material to prevent vermin infestation and corrosion.
   a. If the boot has a screw take-up, clean the acme take-up screws and nuts. Apply a protective coating of rust-inhibiting lubricant. Future take-up adjustment will be much easier.

      (OR)

   b. If the boot has a posi-guide gravity take-up, cleanse the polyethylene guide sleeves and the stainless steel guide shafts of accumulated dirt to ensure that the pulley and weight box assembly move up and down freely. The guide sleeves and shaft **DO NOT** require any lubrication. In fact, any lubricant applied would merely attract dirt, preventing smooth operation of the take-up.
3. Examine the head lining. Extreme wear patterns can sometimes distort the discharge and if not corrected, wear completely through the head. Spouting may also wear through.
4. Check to make sure the pressure relief vents, if installed, on the head and/or legging are unobstructed. In order to provide relief, they must be free to blow out.
5. Inspect all ladders and platforms. Tighten any loose fasteners. Note any defective field welds and schedule immediate repair. Also schedule replacement for any damaged ladder sections, platform structural members, or floor grates. While inspecting platforms and ladders, be sure to examine guying cable brackets and/or bracing. Note any defects and schedule immediate repair.

6. Examine guying. Tighten any clamp fasteners, which are loose. If a cable is excessively slack, adjust turnbuckles to restore tension.



Excessive guy cable tension will twist, bend and/or collapse the elevator structure, causing severe injury or death. Maintain proper guy cable tension at all times.

7. Check the belt take-up in the boot section. If the pulley has been adjusted to its lowest (maximum) limit, it will be necessary to shorten the belt by removing one or two (2) rows of buckets and resplice the belt as explained belt splicing *on Page 33*.
8. It is also advisable to check the condition of all infeed and discharge spouting associated with the elevator for damage and wear.
9. When all specific inspections have been carried out, stand back and generally observe the elevator and any associated equipment. Look and listen for any problem not covered in the preceding list.

Lubrication

In all cases, the manufacturers of the individual components have precise recommendations for periodic lubrication of their products. Strict adherence to these procedures will result in a minimum of down time and maximum component life.

Reducer

Refer to the documentation furnished with the reducer. The user must interpret this information in light of the severity of duty in each application. If there is any doubt, contact the manufacturer or a local supplier of the reducer for specific recommendations.

Motor

Many motors have sealed and permanently lubricated bearings; with these, no re-lubrication is possible or desirable. If bearings of this type become noisy or overheat, they must be replaced.

Motors having bearings which can be re-lubricated are usually larger integral horsepower sizes. Special pressure lubricating equipment may be required. Refer to the documentation furnished with the motor.

Head and Boot Shaft Bearings

Mounted bearings DO require periodic lubrication. The amount and frequency depends in large extent upon the severity of the operating environment and the duty cycle. Refer to manufacturer's recommendations for frequency, type and amount of lubrication.

Roller Chain Drive

If the elevator includes a roller chain drive furnished by GSI, the chain case or guard also serves as a lubricant reservoir. The lubricant level should be high enough to immerse the lower sprocket teeth and chain, yet not so high as to leak from the joints in the case.

Sleeve Guides and Shafts

The polyethylene sleeve guides and stainless steel shafts of posi-guide gravity boot take-ups DO NOT require ANY lubricant. In fact any lubricant applied to these components will only attract dirt and interfere with their operation.

Adjustment and Repair Procedure

Boot Take-Up (Belt Tension) Adjustment

In-service belt tension *on Page 39* states the GSI position regarding proper belt tension. *Step A* of in-service belt tension *on Page 39* explains how to adjust a screw take-up boot or how to add weight to a gravity take-up boot.

Drive V-Belt (If Installed)

If the drive includes V-belts, there are several methods of measuring tension.

1. Direct reading using a gauge specifically designed to read belt tension. Consult a local industrial V-belt supplier.
2. Indirect method using a pull hook scale and a straightedge. The engineering section of many catalogs explains the procedure. In either event, most V-belt catalogs explain proper belt tensioning procedures and give specific tension values.

Belt Tracking

Initial belt tracking *on Page 36* explains the recommended method of tracking the belt.

READ THIS FIRST

If a tracking problem exists and cannot be corrected by adjusting the boot take-up but instead requires that the head and/or boot pulley shaft pillow blocks be shimmed, there may have been a shift in the elevator structure. Look for the following conditions, in order of probability:

1. Worn or damaged pulley
2. Uneven belt stretch. Perhaps the belt is tearing along one edge. Check the splices.
3. Prolonged uneven infeeding of elevated material along one edge of belt or off center loading on the boot.
4. Bent head or boot pulley shaft.
5. Bending or buckling of one or more elevator sections.
6. Damage to guy cable system or deadman shifting.

Belt Splicing

Refer to belt splicing *on Page 33*.

Throat Slide Adjustment

Refer to rubber throat slide adjustment *on Page 38*.

Roller Chain Tension

Many roller chain catalogs include instructions for properly tensioning roller chains and give specific values. Excessive tension can cause accelerated chain and sprocket wear, destroy bearings, and bend shafts. In extreme cases, excessive tension can cause structural damage.

Limited Warranty - N.A. Grain Products

The GSI Group, LLC. (“GSI”) warrants products which it manufactures, to be free of defects in materials and workmanship under normal usage and conditions for a period of 12 months from the date of shipment (or, if shipped by vessel, 14 months from the date of arrival at the port of discharge). If, in GSI’s sole judgment, a product is found to have a defect in materials and/or workmanship, GSI will, at its own option and expense, repair or replace the product or refund the purchase price. This Limited Warranty is subject to extension and other terms as set forth below.

Warranty Enhancements:

The warranty period for the following products is enhanced as shown below and is in lieu of (and not in addition to) the above stated warranty period. (Warranty Period is from date of shipment.)

	Product	Warranty Period
Storage	Grain Bin Structural Design • Sidewall, roof, doors, platforms and walkarounds • Flooring (when installed using GSI specified floor support system for that floor) • Hopper tanks (BFT, GHT, NCHTC, and FCHTC)	5 Years
Conditioning	Dryer Structural Design - (Tower, Portable and TopDry) • Includes (frame, portable dryer screens, ladders, access doors and platforms)	5 Years
	All other Dryer parts including: • Electrical (controls, sensors, switches and internal wiring)	2 Years
	All Non-PTO Driven Centrifugal and Axial Fans	3 Years
	GrainVue	2 Years
Material Handling	Bucket Elevators Structural Design	5 Years
	Towers Structural Design	5 Years
	Catwalks Structural Design	5 Years
	Accessories (stairs, ladders and platforms) Structural Design	5 Years

Conditions and Limitations:

THERE ARE NO WARRANTIES THAT EXTEND BEYOND THE LIMITED WARRANTY DESCRIPTION SET FORTH HEREIN; SPECIFICALLY, GSI DISCLAIMS ANY AND ALL OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING, WITHOUT LIMITATION, WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE OR USE IN CONNECTION WITH: (I) ANY PRODUCT MANUFACTURED OR SOLD BY GSI, OR (II) ANY ADVICE, INSTRUCTION, RECOMMENDATION OR SUGGESTION PROVIDED BY AN AGENT, REPRESENTATIVE OR EMPLOYEE OF GSI REGARDING OR RELATED TO THE CONFIGURATION, INSTALLATION, LAYOUT, SUITABILITY FOR A PARTICULAR PURPOSE, OR DESIGN OF SUCH PRODUCTS.

The sole and exclusive remedy for any claimant is set forth in this Limited Warranty and shall not exceed the amount paid for the product purchased. This Warranty only covers the value of the warranted parts and equipment, and does not cover labor charges for removing or installing defective parts, shipping charges with respect to such parts, any applicable sales or other taxes, or any other charges or expenses not specified in this Warranty. GSI shall not be liable for any other direct, indirect, incidental or consequential damages, including, without limitation, loss of anticipated profits or benefits. Expenses incurred by or on behalf of a claimant without prior written authorization from the GSI warranty department shall not be reimbursed. This warranty is not transferable and applies only to the original end-user. GSI shall have no obligation or responsibility for any representations or warranties made by or on behalf of any dealer, agent or distributor. Prior to installation, the end-user bears all responsibility to comply with federal, state and local codes which apply to the location and installation of the products.

This Limited Warranty extends solely to products sold by GSI and does not cover any parts, components or materials used in conjunction with the product, that are not sold by GSI. GSI assumes no responsibility for claims resulting from construction defects, unauthorized modifications, corrosion or other cosmetic issues caused by storage, application or environmental conditions. Modifications to products not specifically delineated in the manual accompanying the product at initial sale will void all warranties. This Limited Warranty shall not extend to products or parts which have been damaged by negligent use, misuse, alteration, accident or which have been improperly/inadequately maintained.

Notice Procedure:

In order to make a valid warranty claim a written notice of the claim must be submitted, using the RMA form, within 60 days of discovery of a warrantable nonconformance. The RMA form is available on Dealer Connect.

Service Parts:

GSI warrants, subject to all other conditions described in this Warranty, Service Parts which it manufactures for a period of 12 months from the date of purchase unless specified in Enhancements above.

(Limited Warranty - N.A. Grain Products_revised 22 May 2026)

This equipment shall be installed in accordance with the current installation codes and applicable regulations, which should be carefully followed in all cases. Authorities having jurisdiction should be consulted before installations are made.

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